

Introduction to Earth System

Oscillations

1. El Niño Southern Oscillation (ENSO)
2. Quasi-biennial Oscillation (QBO)
3. Madden-Julian Oscillation (MJO)

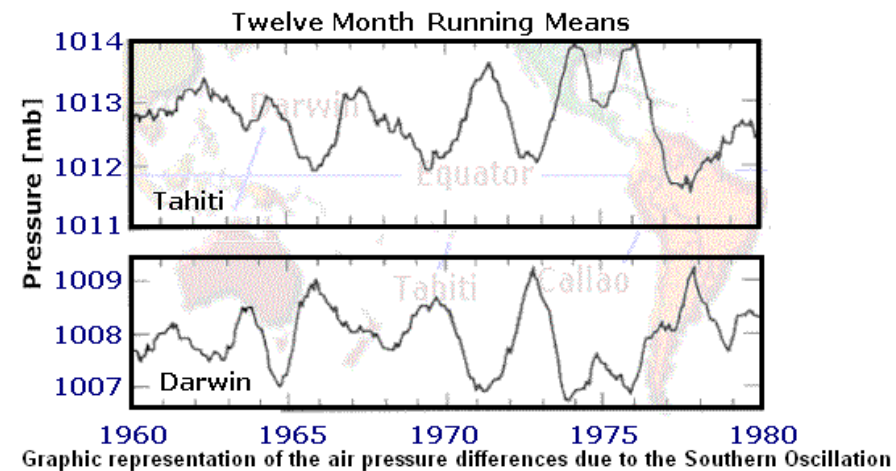
I/ ENSO

(El Niño Southern Oscillation)

1. The concept of ENSO,
2. Physical mechanism, characteristics
3. ENSO index
4. Impact of ENSO

ENSO

- ❑ **El Niño**: a large-scale climate phenomena linked to the **periodic warming of SST** across the central and east-central equatorial Pacific.
- ❑ **La Niña**: the **periodic cooling of SST**
- ❑ **ENSO**: El Niño Southern Oscillation → El Niño, La Niña accompanying atmospheric component, i.e. air pressure oscillation in the east and west pacific-east Indian ocean (Southern Oscillation)

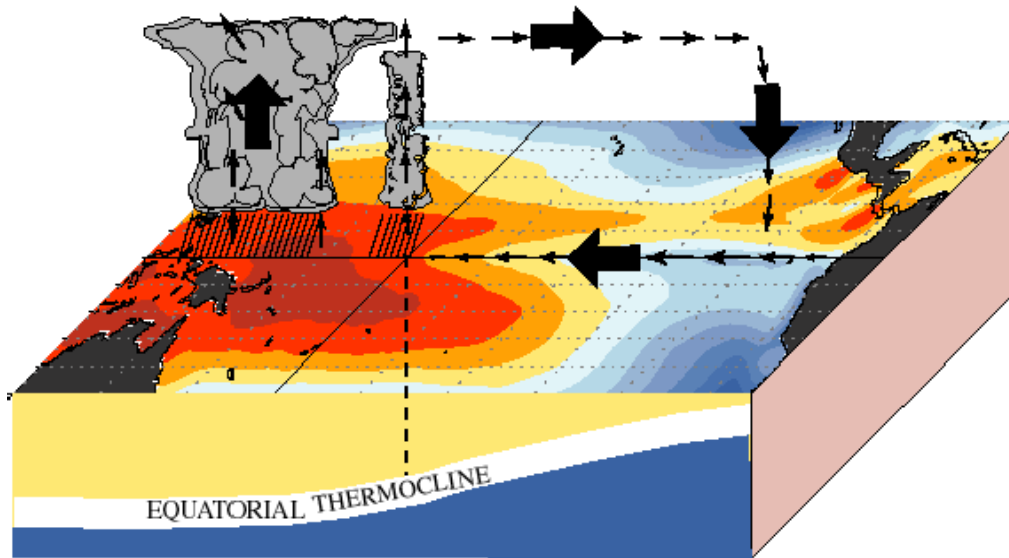


Walker identified the **sea-saw oscillation** from the air pressure data (1920s) obtained in Tahiti & Darwin → **Southern Oscillation**

ENSO

Jacob Bjerknes (1966) links Southern Oscillation with El Niño

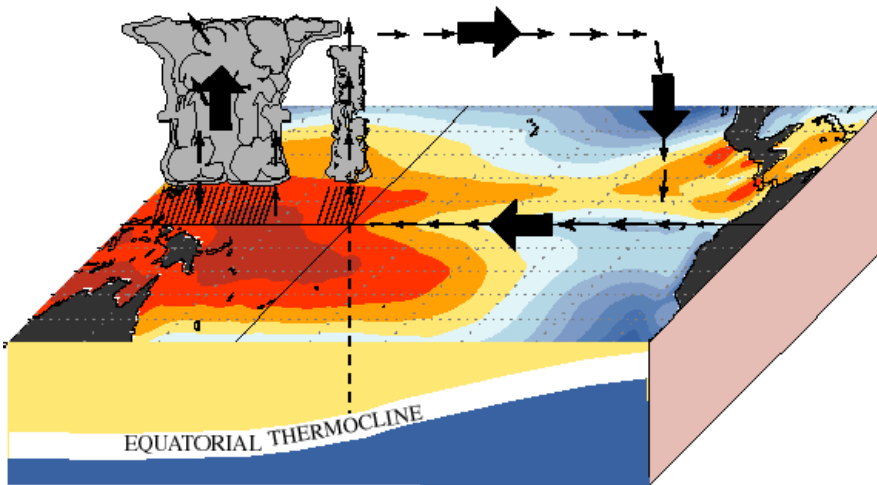
December - February Normal Conditions



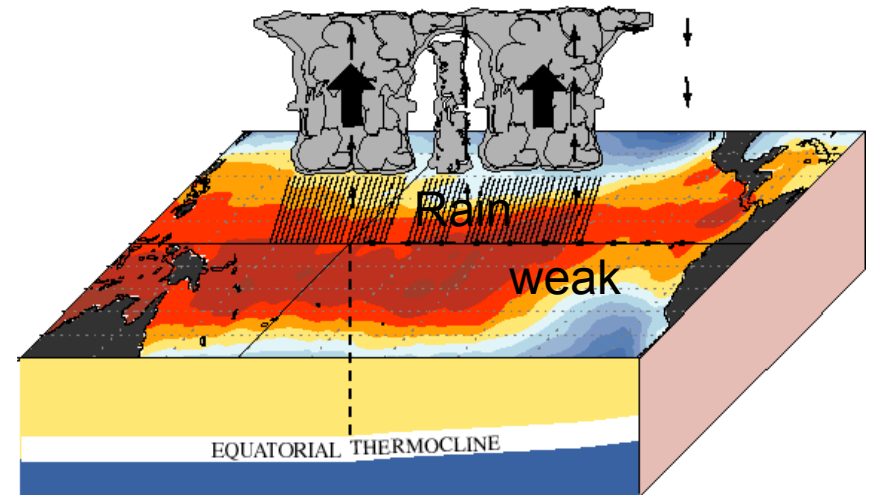
- difference between East & West
 - Pressure?
 - Temperature?
 - easterly wind at low level and westerly wind at high altitude;
 - **downdraft in East** and **updraft in West** → close circulation: **Walker Cell**

- Differences b.w. East & West increase → stronger Walker circulation → the thermocline inclines more

December - February Normal Conditions



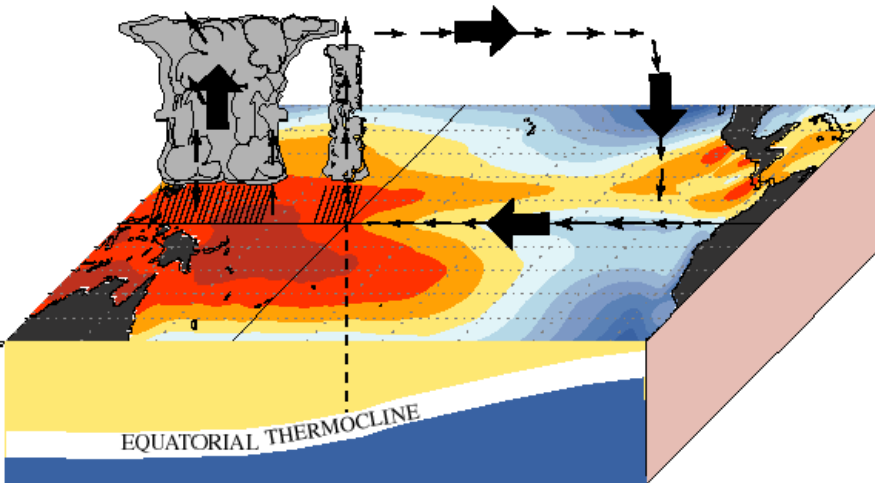
December - February El Niño Conditions



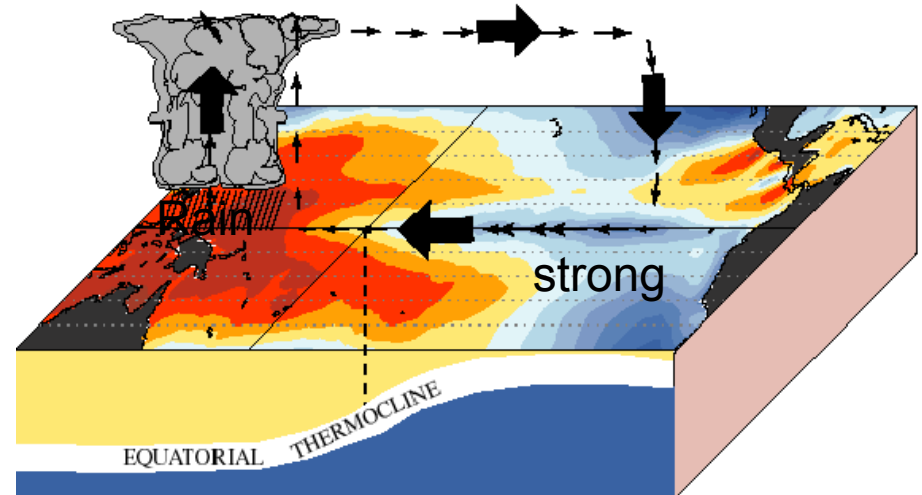
El Niño

- Walker Cell weakens
- Weakening of easterlies & upwelling
- → the eastern Pacific experiences a warm SST anomaly
→ more convection over the eastern Pacific;
- The thermocline inclines less

December - February Normal Conditions

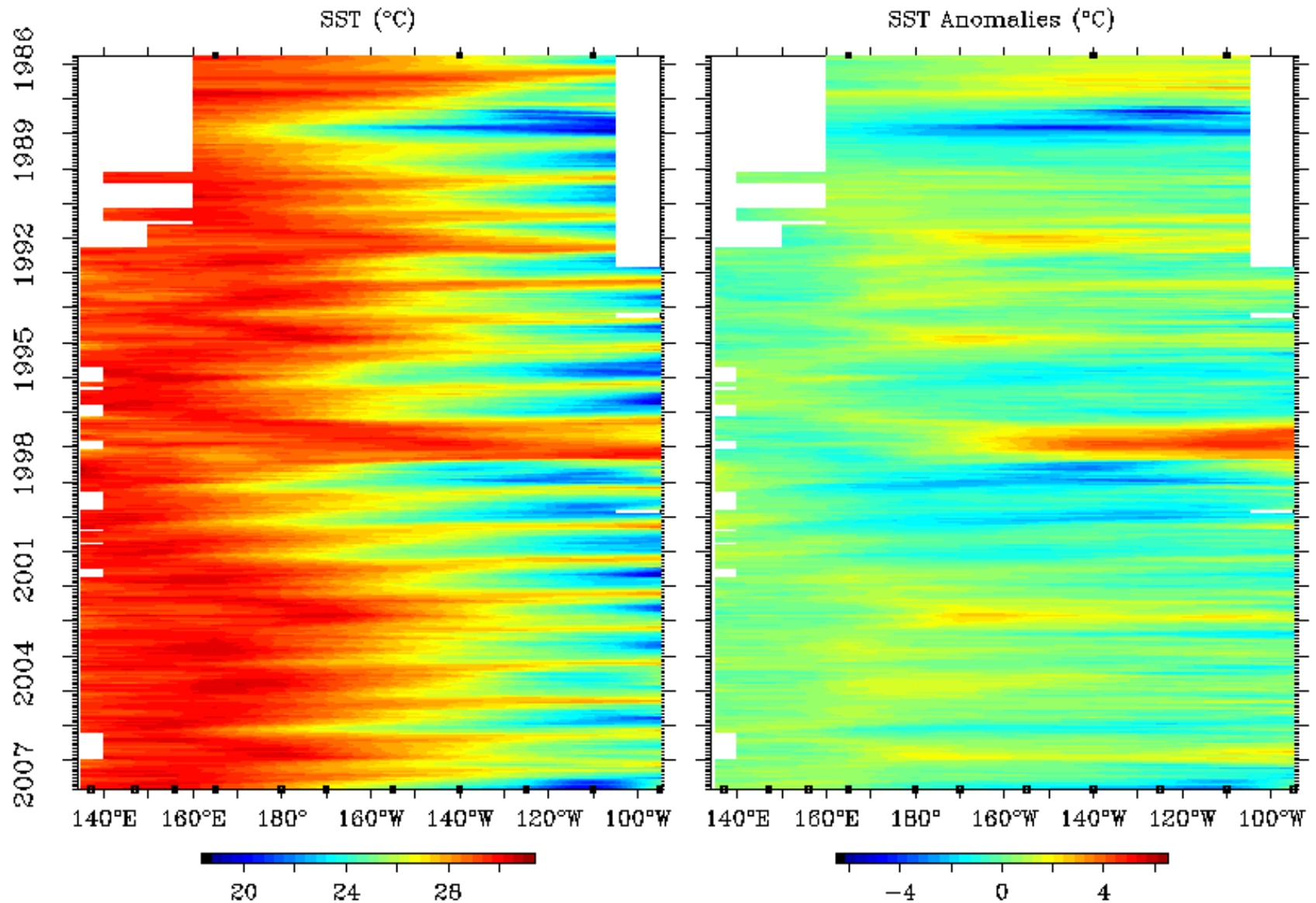


December - February La Niña Conditions

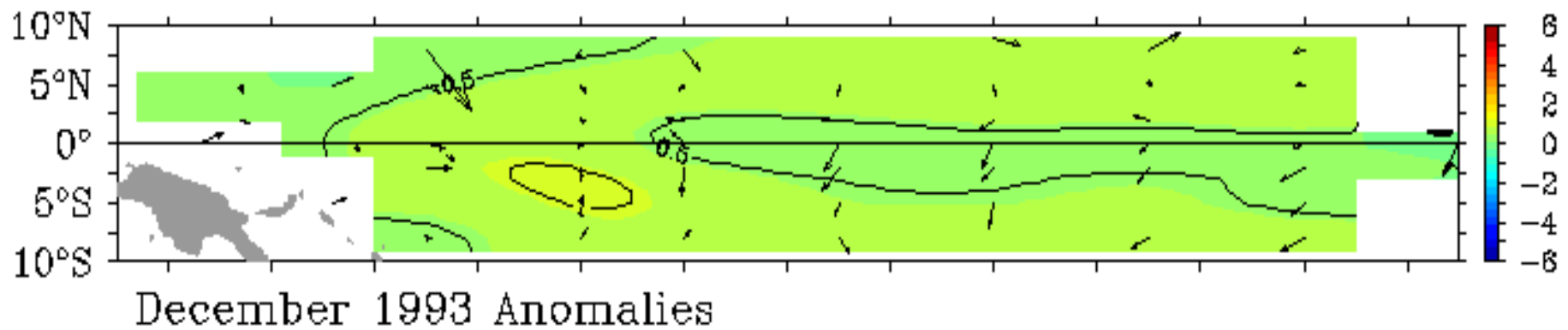
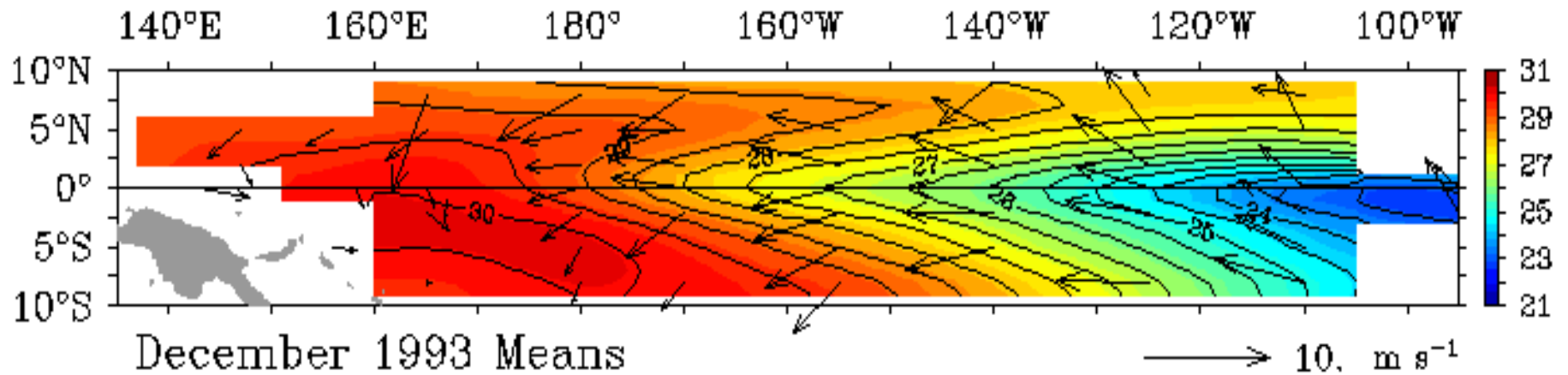


La Niña → the opposite phase

Five-Day SST 2°S to 2°N Average



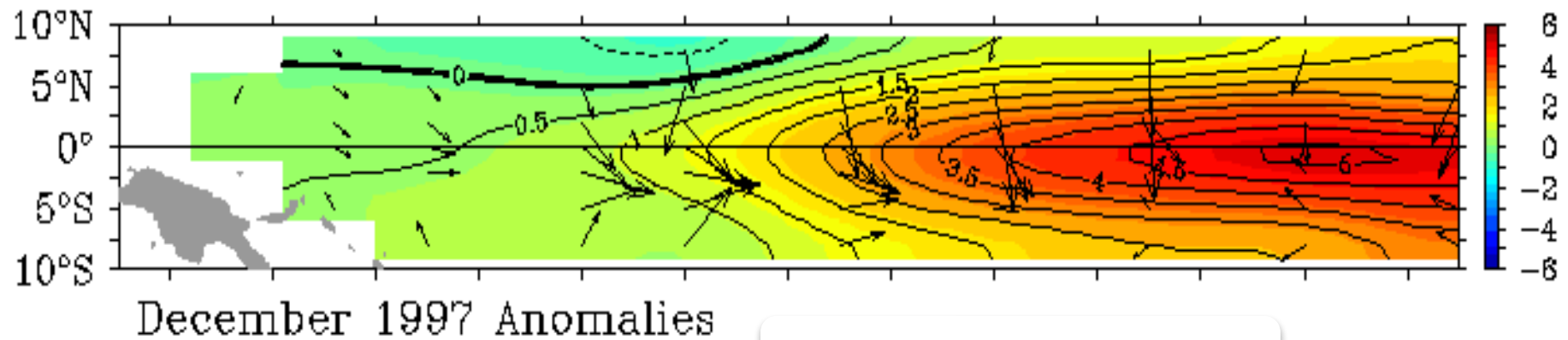
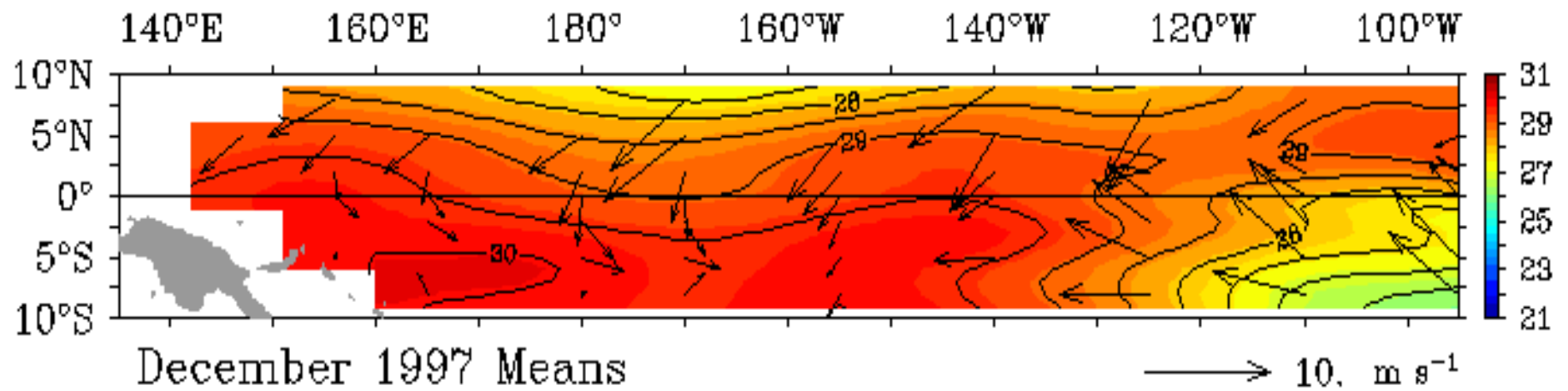
TAO Monthly Mean SST ($^{\circ}\text{C}$) and Winds (m s^{-1})



TAO Project Office/PMEL/NOAA

Q. El niño or La niña

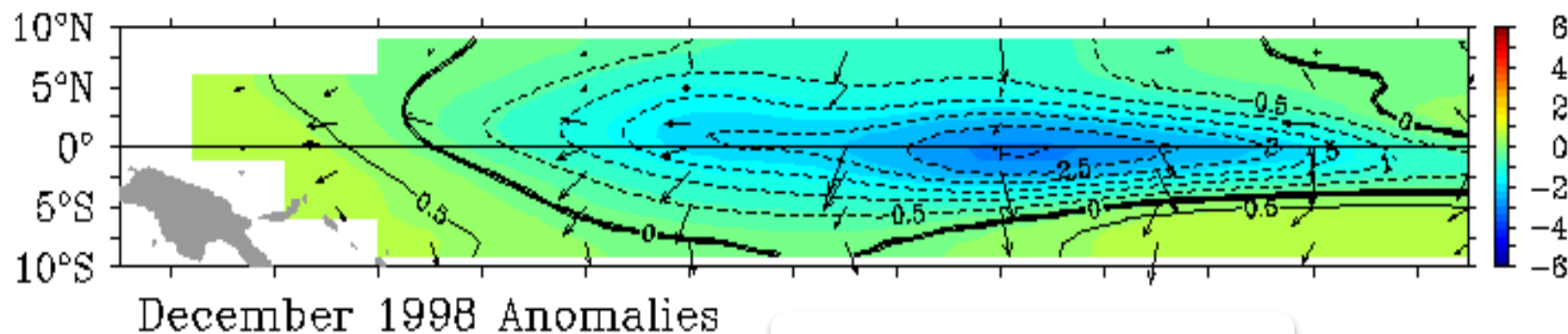
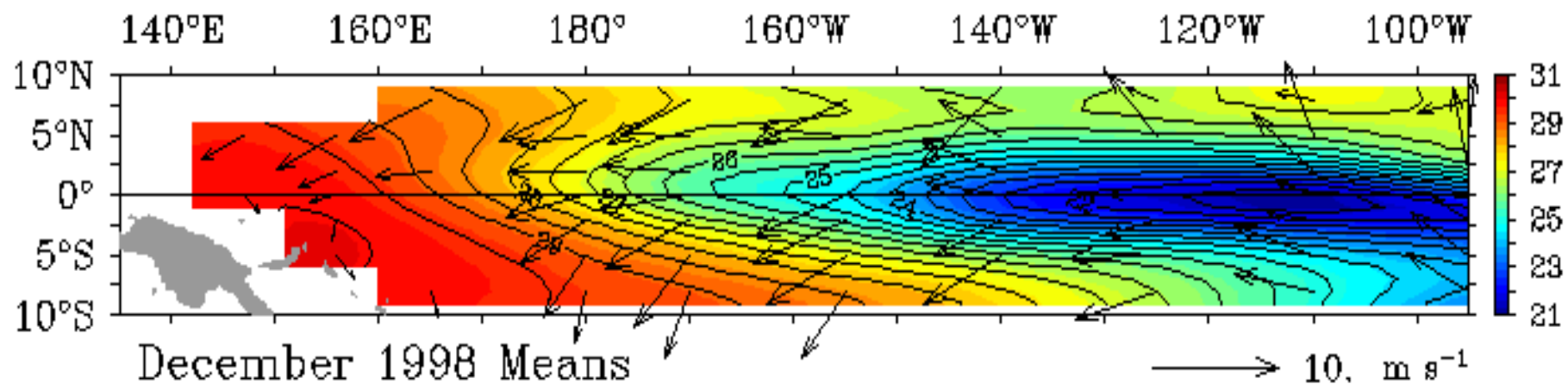
TAO Monthly Mean SST ($^{\circ}\text{C}$) and Winds (m s^{-1})



TAO Project Office/PMEL/NOAA

Q. El niño or La niña

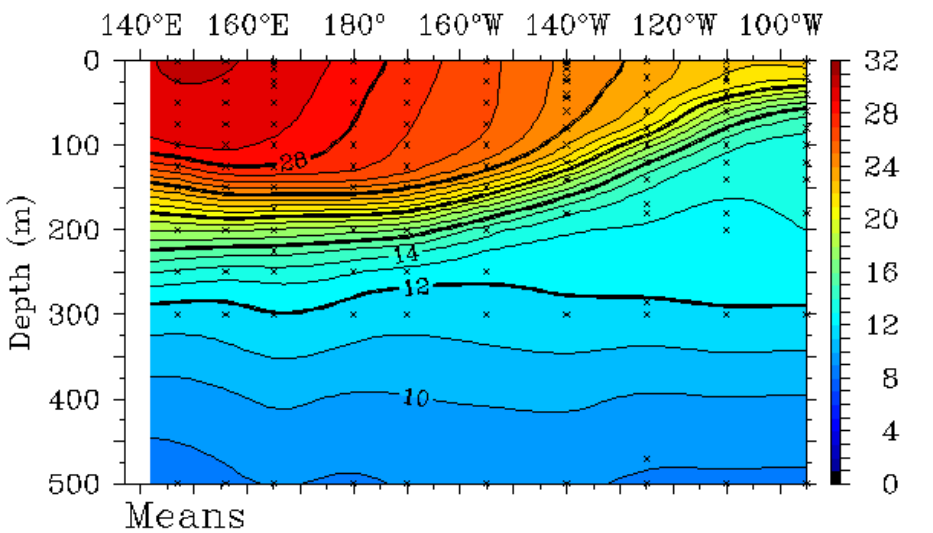
TAO Monthly Mean SST ($^{\circ}\text{C}$) and Winds (m s^{-1})



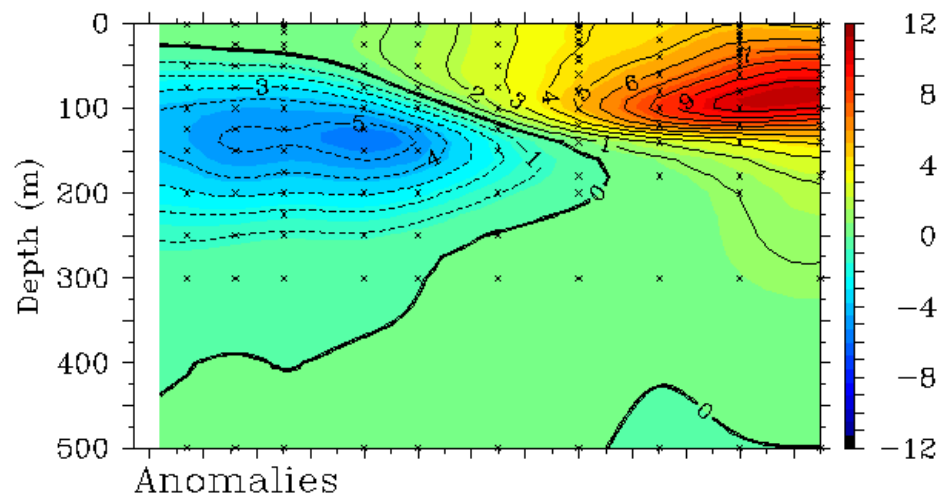
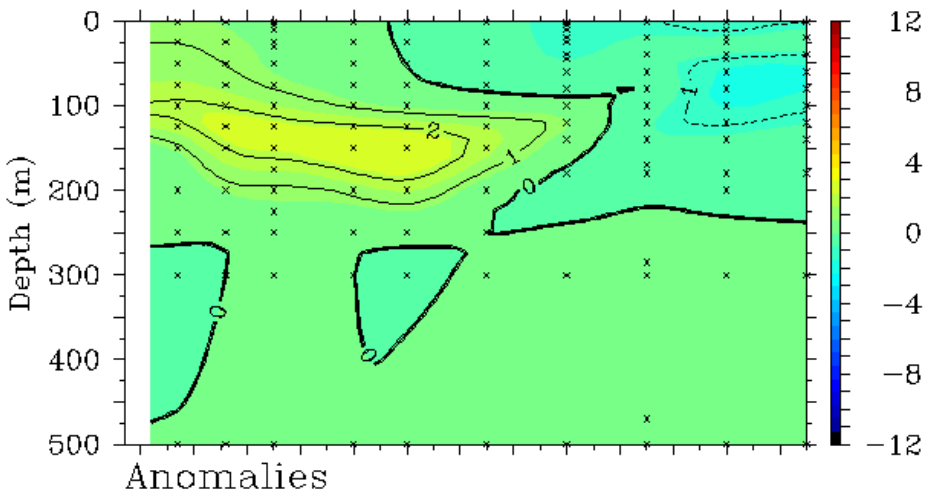
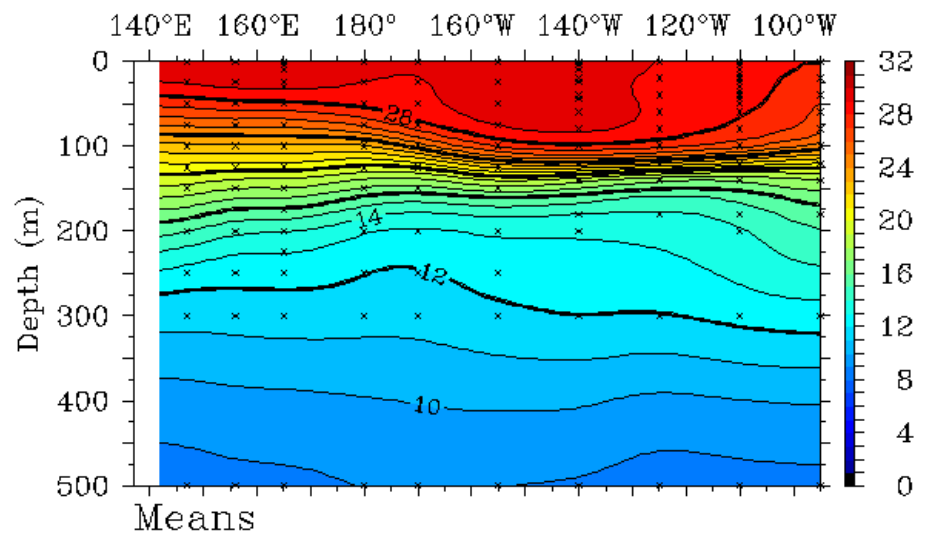
TAO Project Office/PMEL/NOAA

Q14. El niño or La niña

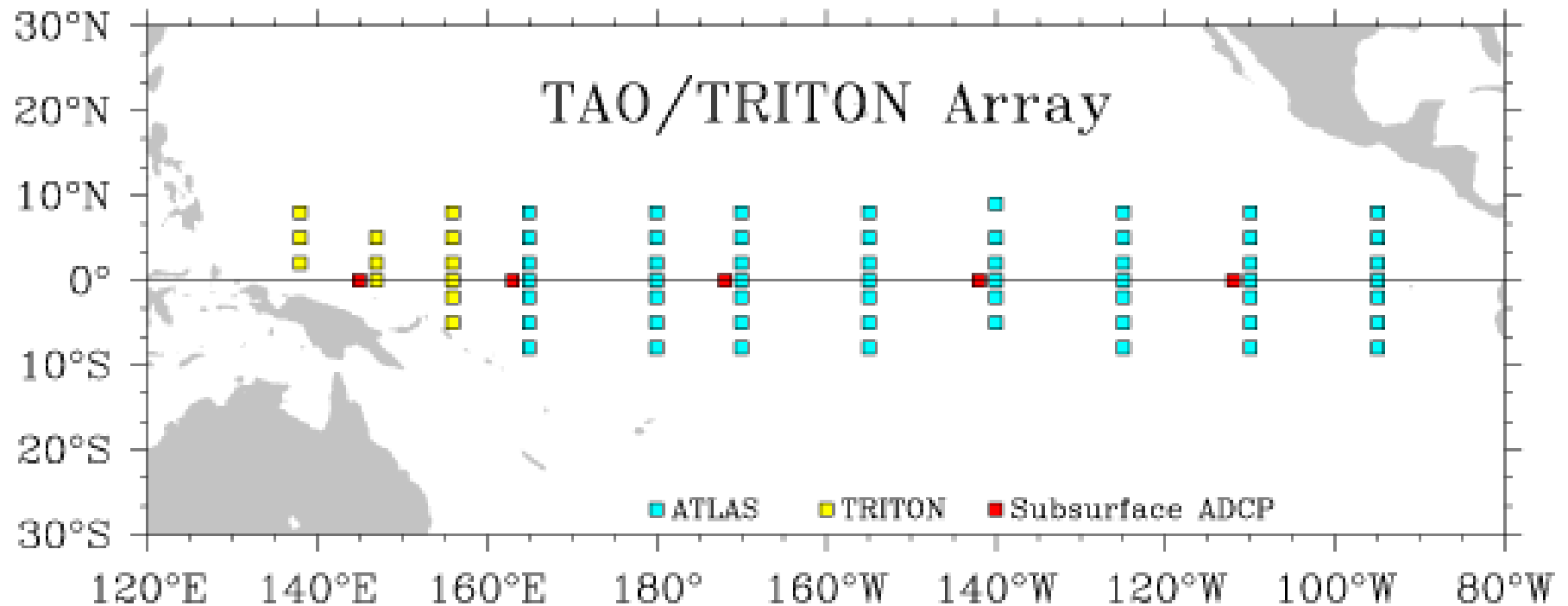
Monthly Mean TAO/TRITON Temperatures (°C)
December 1996 2°S to 2°N Average



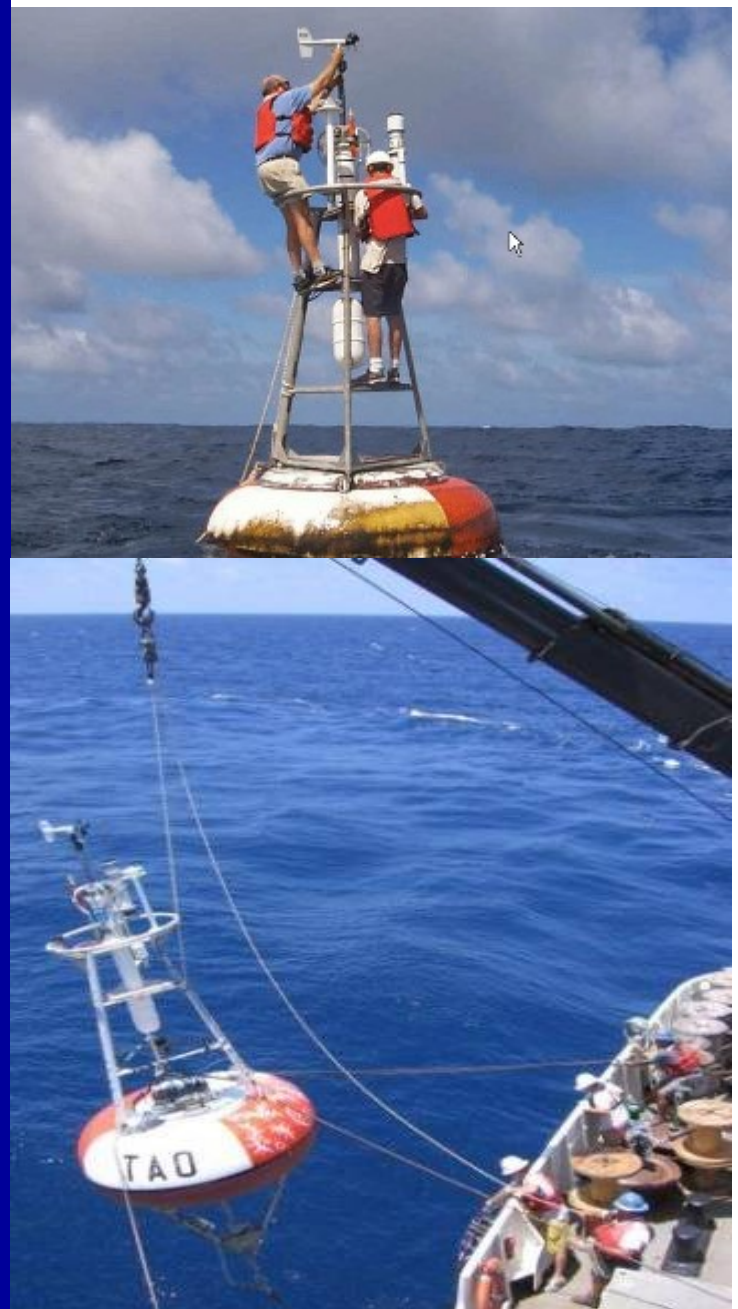
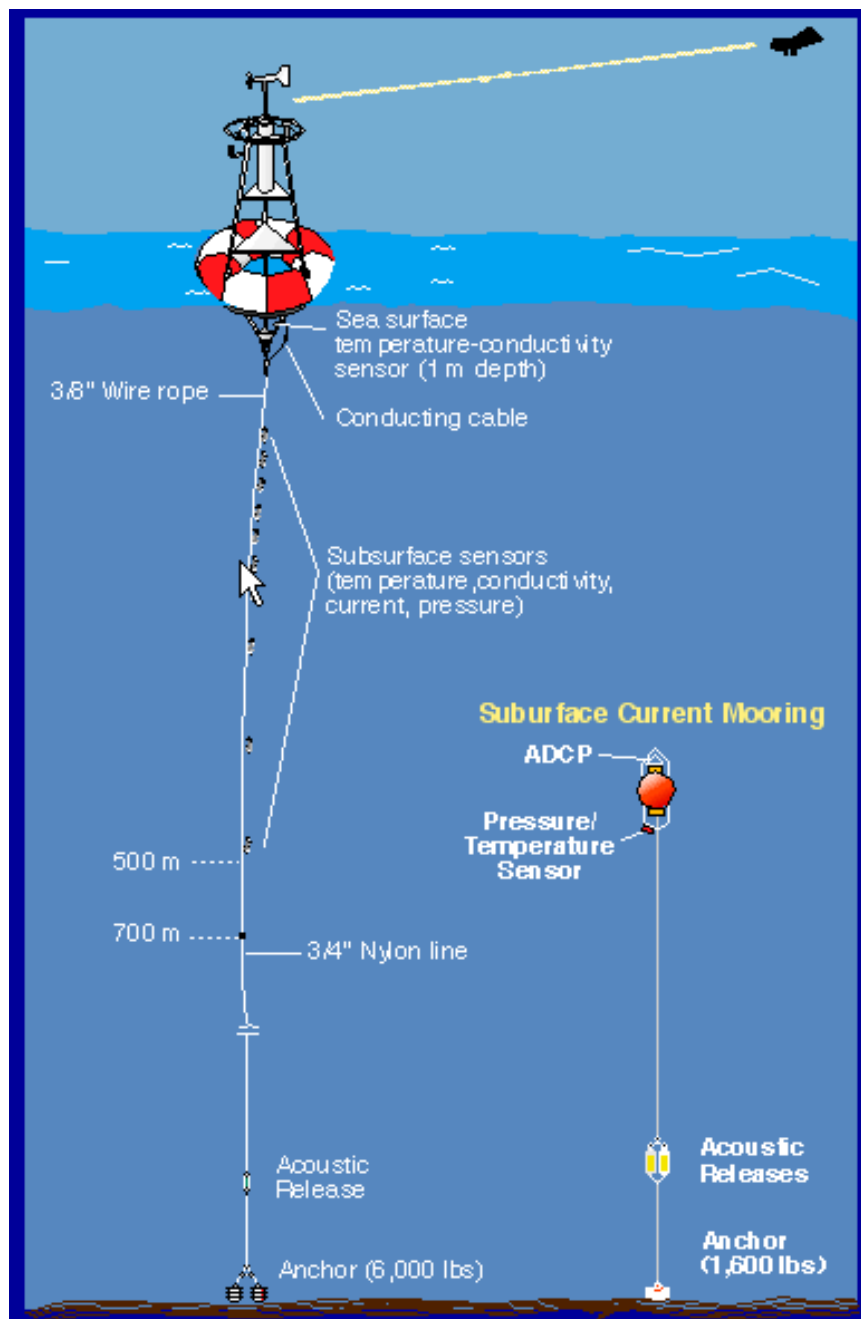
Monthly Mean TAO/TRITON Temperatures (°C)
December 1997 2°S to 2°N Average



ENSO index using SST

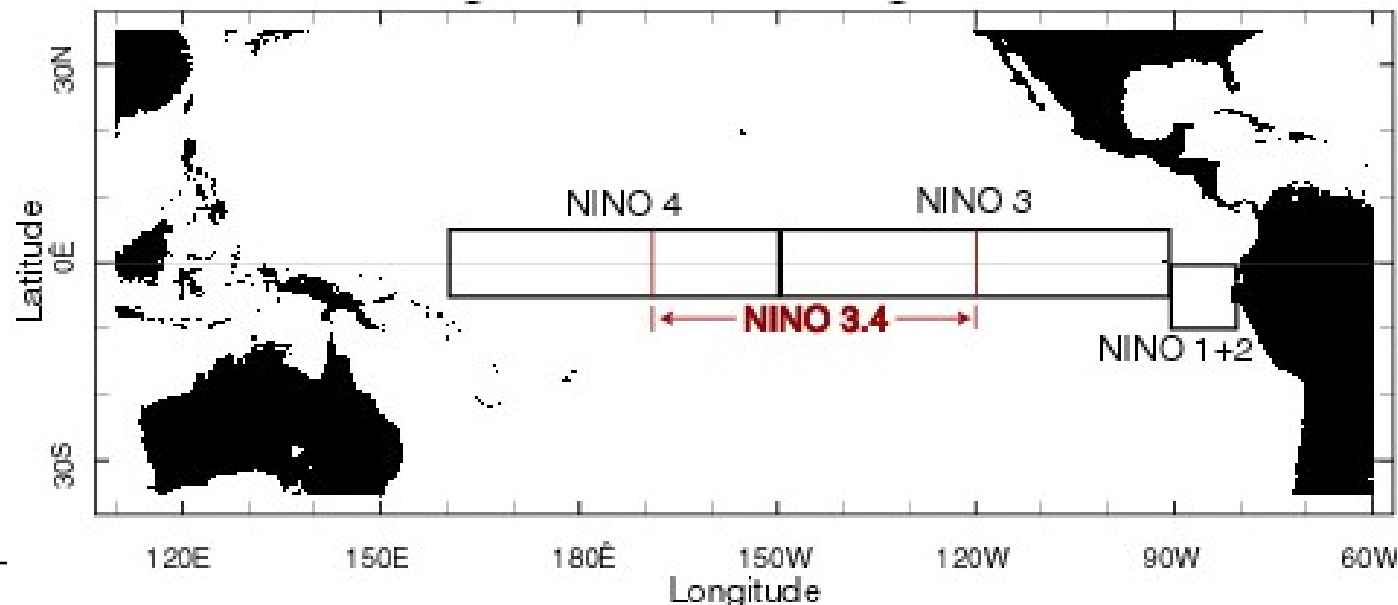


Picture courtesy of the Pacific Marine Environmental Laboratory (PMEL)



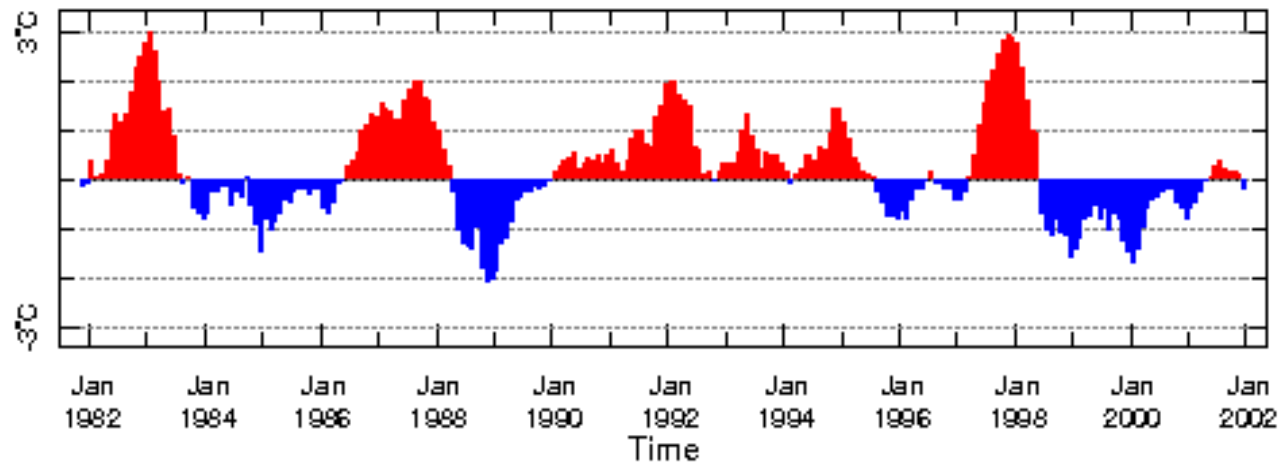
NIÑO index regions

- **NIÑO1+2** (0-10S, 80-90W): the first warming region when El Niño develops
- **NIÑO3** (5S-5N; 150W-90W): region where SST changes are the highest during El Niño
- **NIÑO3.4** (5S-5N; 170W-120W): region where SST changes are important for the shift of rainfall area.
- **NIÑO4** (5S-5N; 160E-150W): could be good for La Nina detection as an SST anomaly of -0.5C in that region would result in a significant westward shift in the pattern of deep convection in the tropical Pacific.

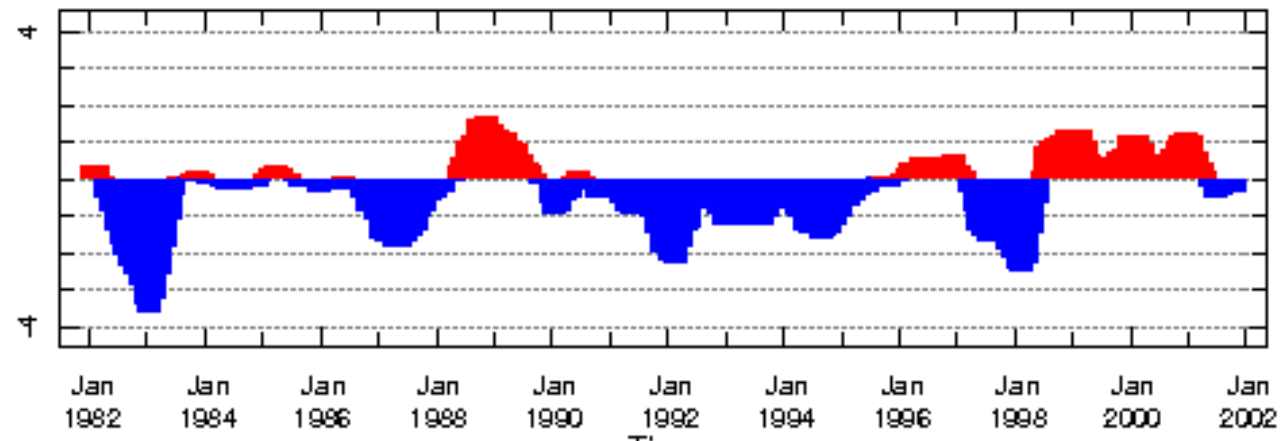


NINO3.4

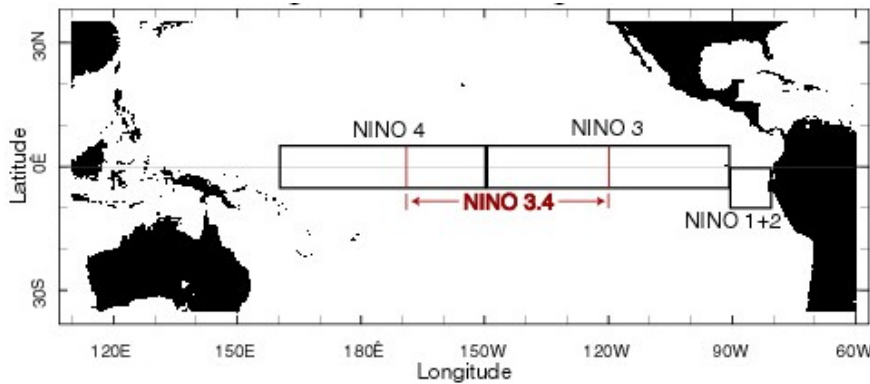
Nino 3.4 Sea Surface Temperature Index (departure from average)



The Southern Oscillation Index



Example: NIÑO3



NIÑO3 (5S-5N; 150W-90W)

- SST: sea surface temperature
- Monthly average SST over the NIÑO3 region: $\{T_{imn, iyr}\}$, *imn* indicates a month (Jan to Dec), *iyr* indicates a year (e.g. 1979, 1980, ..., 2019)
- *SSTA*: Anomaly of SST

$$SSTA_{i,iyr} = T_{i,iyr} - \text{mean}(T_i)$$

where $\text{mean}(T_i)$ is the average of T_i for the chosen period (e.g. 1979-2019)

- El Niño: $SSTA_{5 \text{ months cont.}} \geq 0,5^{\circ}\text{C}$
- La Niña: $SSTA_{5 \text{ months cont.}} \leq -0,5^{\circ}\text{C}$

<ftp://ftp.cdc.noaa.gov/Datasets/noaa.oisst.v2/sst.mnmean.nc>

Example: NIÑO3

- El Niño: $SSTA_{5 \text{ months cont.}} \geq 0,5^{\circ}\text{C}$
- La Niña: $SSTA_{5 \text{ months cont.}} \leq -0,5^{\circ}\text{C}$

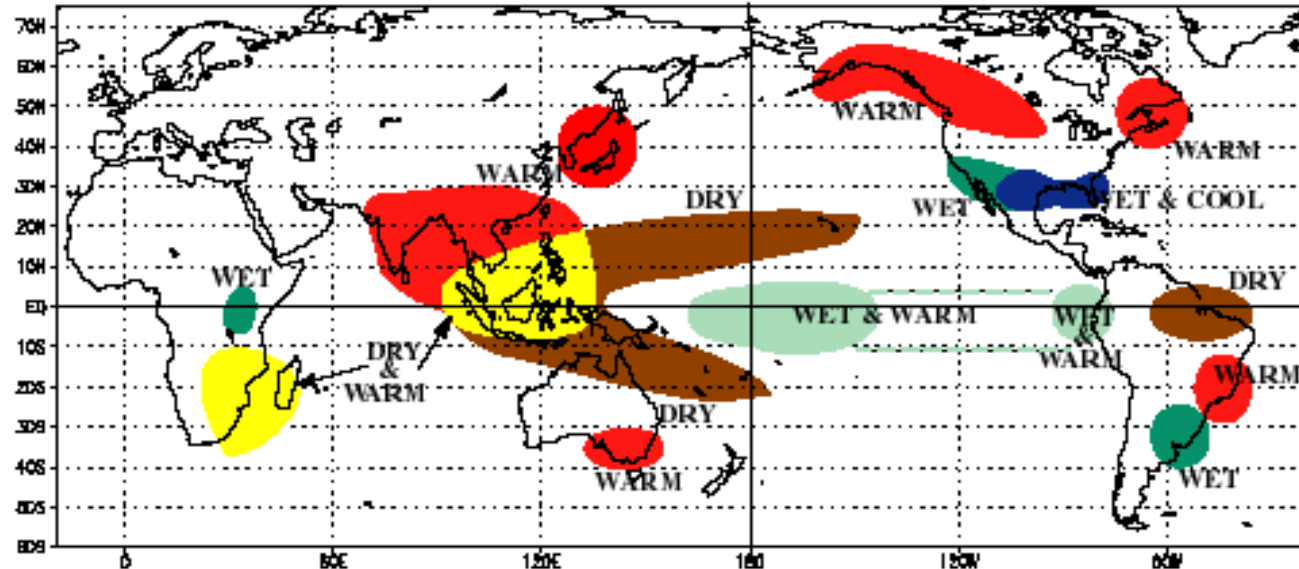
No	El Niño	Duration	Max SSTA (°C) & month that occurs	
1	1951/1952	8	1.3	10/1951
2	1953	9	1.1	9/1953
3	1957/1958	14	1.8	12/1957
4	1963/1964	9	1.2	12/0963
5	1965/1966	10	1.8	12/1965
6	1968/69/70	18	1.4	12/1969
7	1972/1973	12	2.6	12/1972
8	1976/1977	9	1.2	9,10/1976
9	1979	6	1.2	9/1979
10	1982/1983	18	3.6	1/1983
11	1986/87/88	17	2.0	9/1987
12	1991/1992	15	1.7	1/1992
13	1993	7	1.5	5/1993
14	1997/1998	15	3.9	12/1997
15	2002/2003	7	1.4	11,12/2002

No	La Niña	Duration	Max SSTA (°C) & month that occurs	
1	1949/1950	-	-1.7	2/1950
2	1954/55/56	22	-2.0	11/1955
3	1964/1965	10	-1.2	12/1964
4	1967/1968	8	-1.3	2/1968
5	1970/1971	19	-1.5	12/1970
6	1973/1974	10	-1.4	1/1974
7	1975/1976	12	-1.5	12/1975, 1/1976
8	1984/1985	15	-1.2	12/1984
9	1988/1989	12	-1.7	11, 12/1988
10	1998/99/00	18	-1.6	1/2000

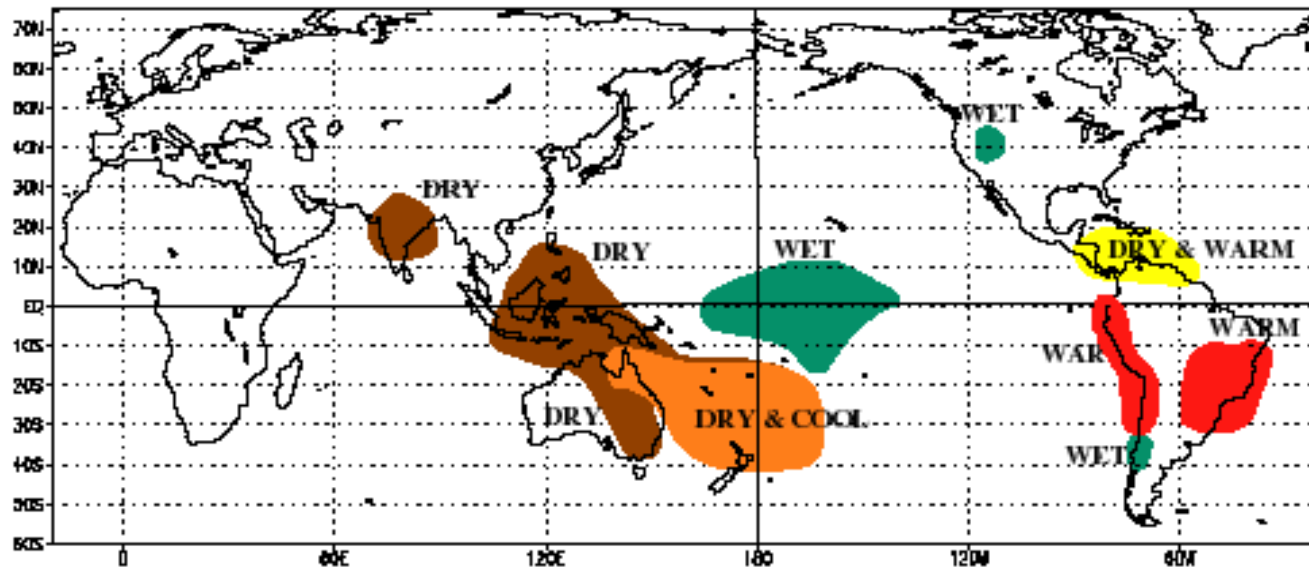
Impacts of ENSO

El Niño

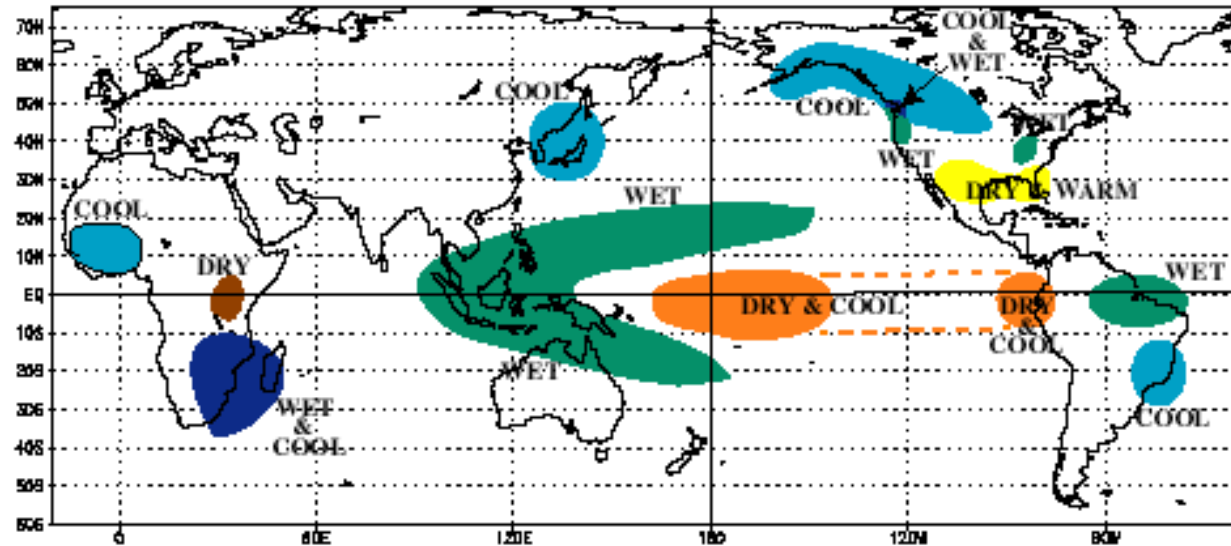
WARM EPISODE RELATIONSHIPS DECEMBER - FEBRUARY



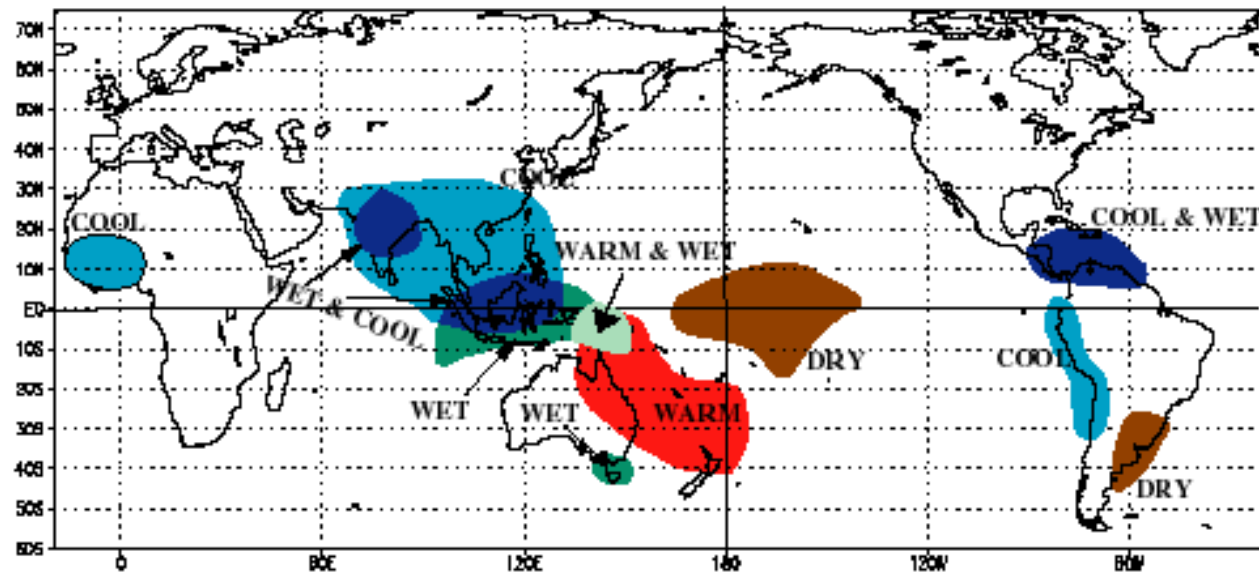
WARM EPISODE RELATIONSHIPS JUNE - AUGUST



COLD EPISODE RELATIONSHIPS DECEMBER - FEBRUARY



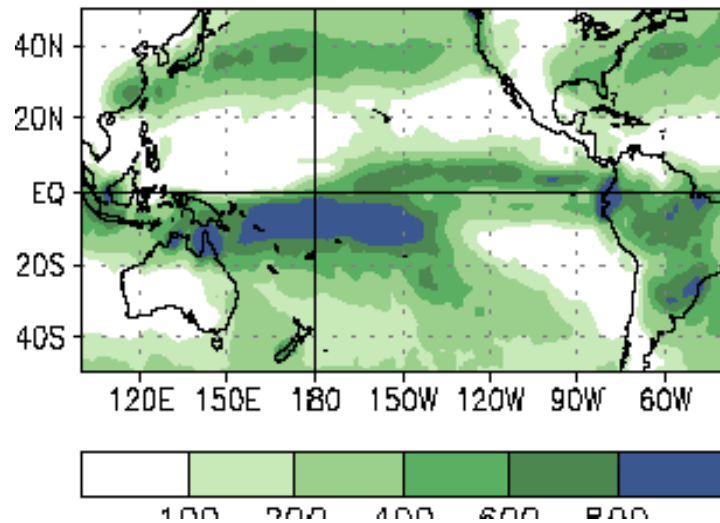
COLD EPISODE RELATIONSHIPS JUNE - AUGUST



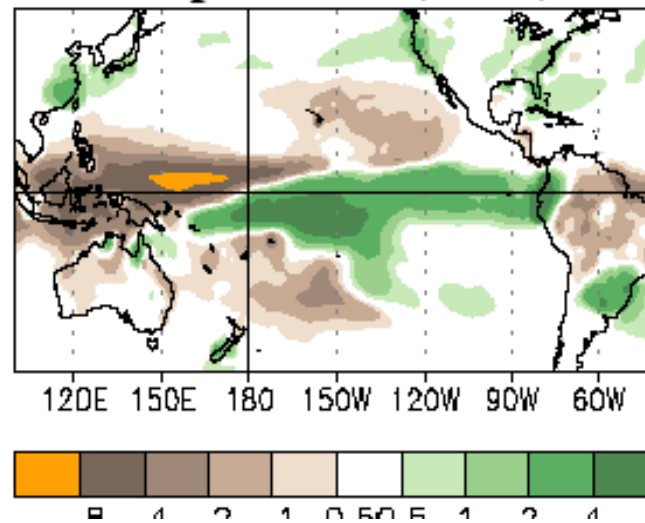
La Niña

Jan-Mar 1998 Precipitation (mm)

Total



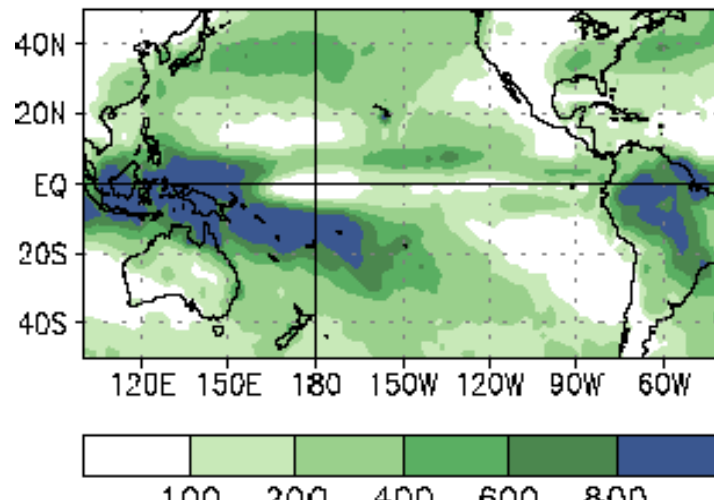
Departures (x100)



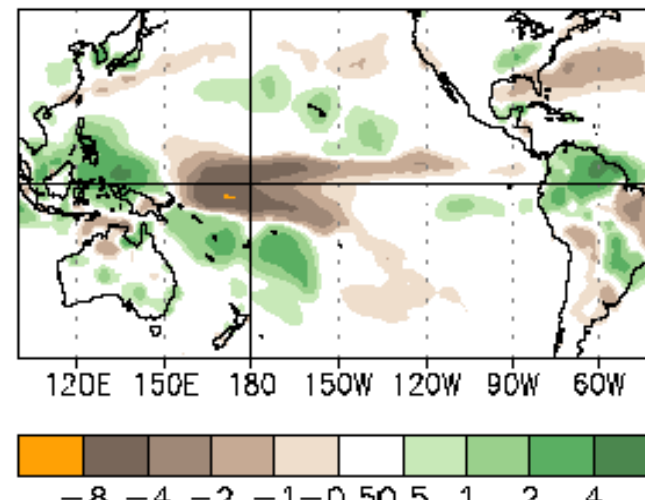
Q. El Niño or
La Niña?

Jan-Mar 1989 Precipitation (mm)

Total

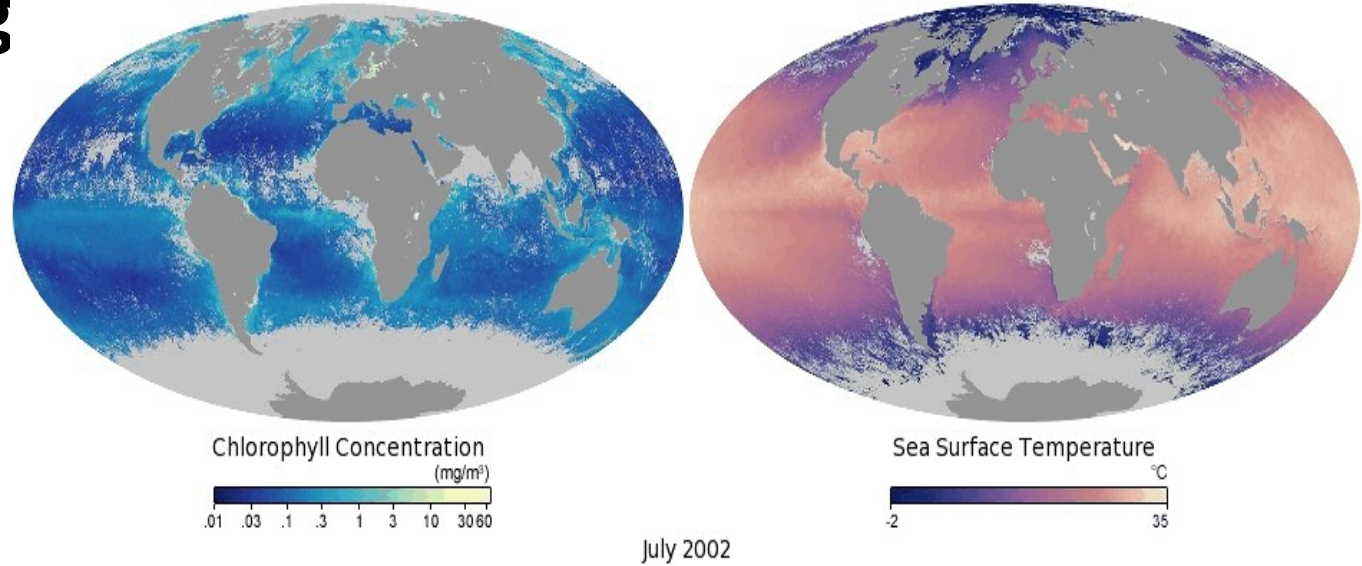


Departures (x100)

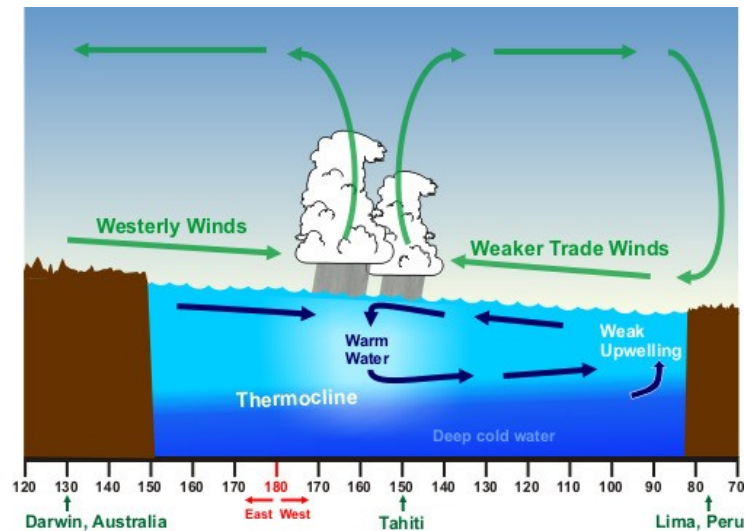


Q. El Niño or
La Niña?

Upwelling



Q: El Niño → Weaker or stronger Upwelling?

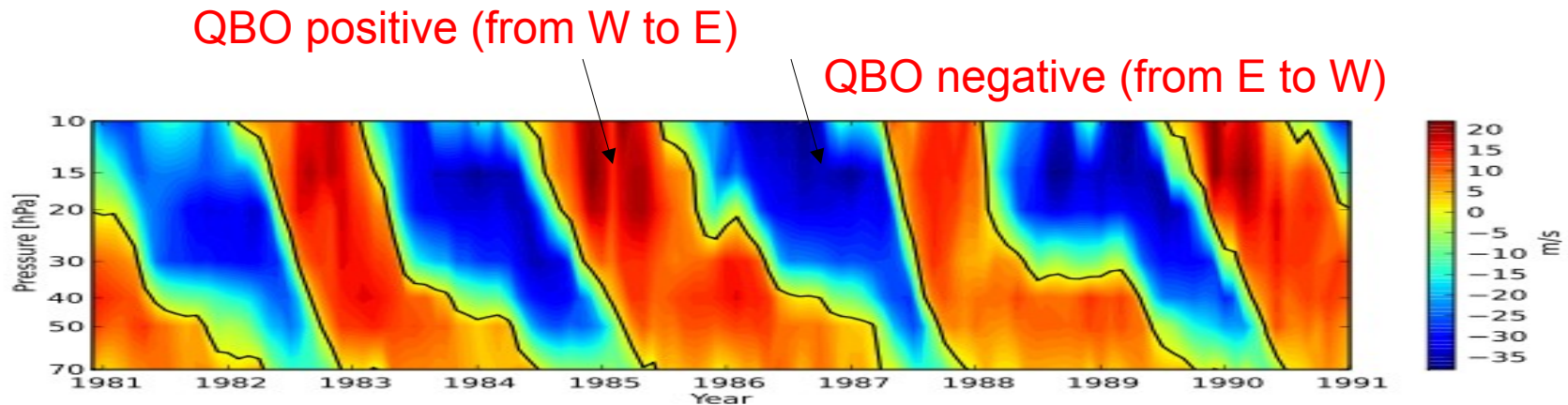


El Niño

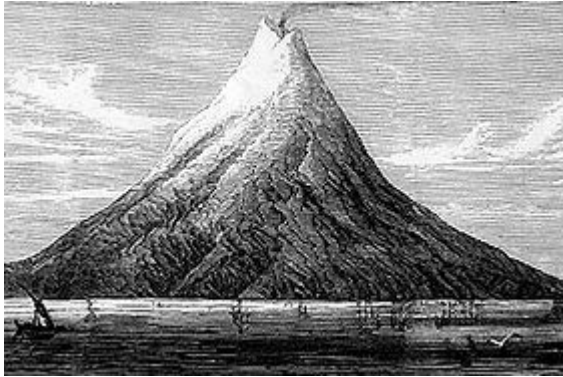
II. Quasi-Biennial Oscillation (QBO)

A phenomenon in which:

- **High-altitude tropical** winds alternate between eastward and westward **directions**
- repeating over a period of almost two years (28 -29 months).



QBO history



- The eruption of Krakatau volcano on August 27, 1883
→ volcanic **dust in the stratosphere blown to the west**, ~13 days to travel around the equator.
- Energy of 13,000 times the atomic bomb dropped in Hiroshima, Japan during World War II, 36417 deaths
→ the most terrible eruption recorded in history
- In 1908, Berson launched a balloon near Lake Victoria in Africa and observed **winds from the west at about 15km** (120hPa). This wind is called Berson West wind

→ **Hypothesis:**

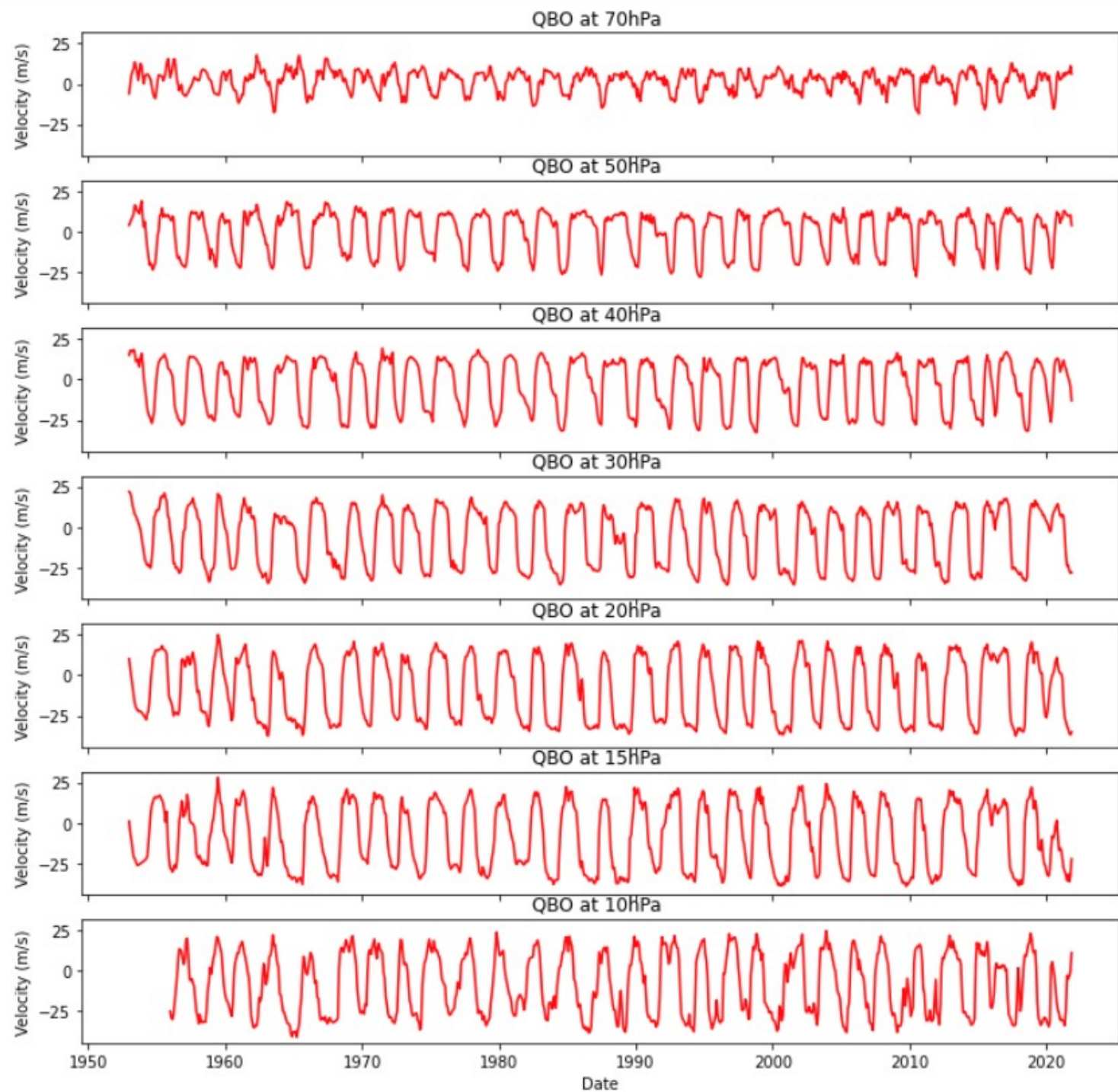
Krakatau easterlies at ~**30km** (10hPa) &
Berson's westerlies at ~**20 km** (50hPa).



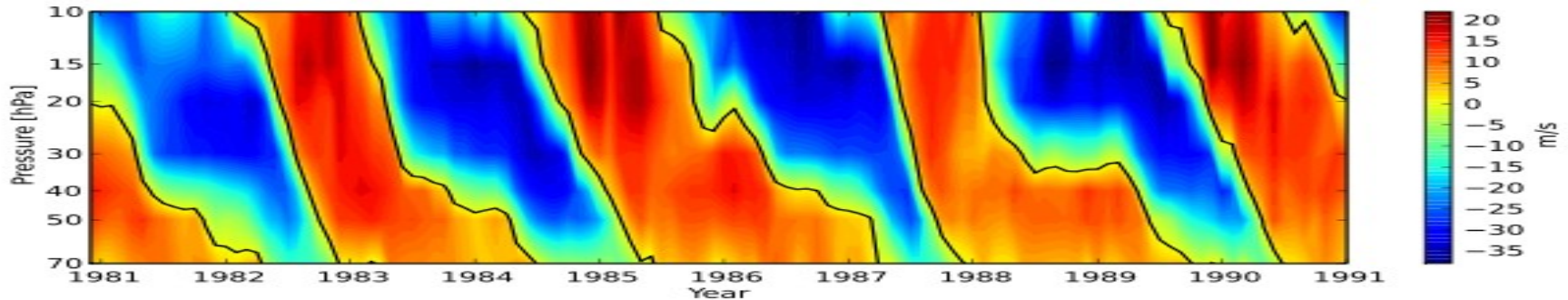
19th century painting about the volcano eruption in 1883

- **Reed and ccs. (1961)**: using wind data around the equator
→ found that from July 1955 to February 1960, high-altitude **zonal winds oscillate every ~ 2 yrs**, peaking near 25 km altitude
 - **Angell & Korshover (1964)**: indicated that periodic fluctuations are also found when considering **temperature, ozone, & tropopause's height**
- QBO concept: quasi-biennial oscillation

- ✓ Reed, R. J., W. J. Campbell, L. A. Rasmussen, and D. G. Rogers (1961), Evidence of a Downward-Propagating, Annual Wind Reversal in the Equatorial Stratosphere, *J. Geophys. Res.*, 66(3), 813–818, doi:10.1029/JZ066i003p00813
- ✓ Angell, J. K., J. Korshover, 1964: Quasi-biennial variations in temperature, total ozone, and tropopause height. *J. Atmos. Sci.*, **21**, 479-492

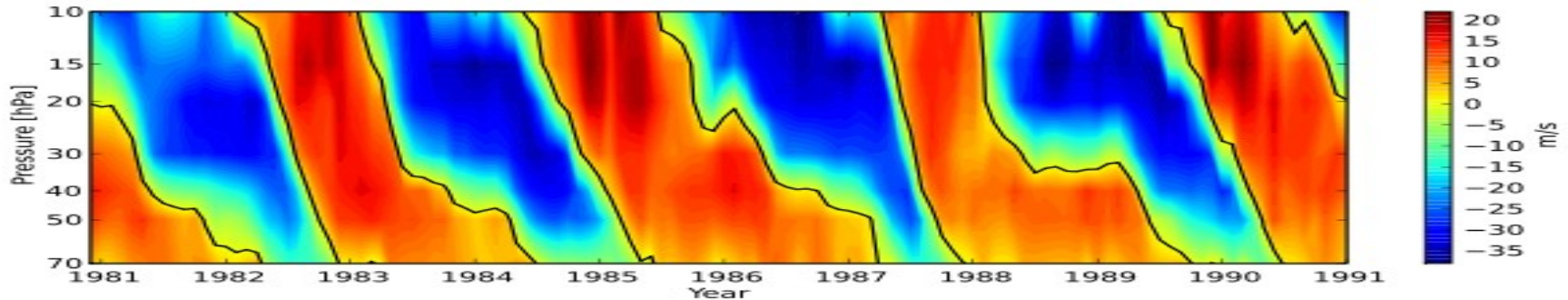


QBO's characteristics



1. Periods range from 20 to 36 months, averaging 28
2. Easterlies are typically twice as strong as westerlies
3. The QBO signal propagates downward over time from $\sim 10\text{hPa}$ ($\sim 30\text{km}$) to 100hPa ($\sim 16\text{km}$) or lower
4. The downward propagating speed is $\sim 1\text{km/month}$
5. Easterlies dominate aloft, while westerlies prevail lower down

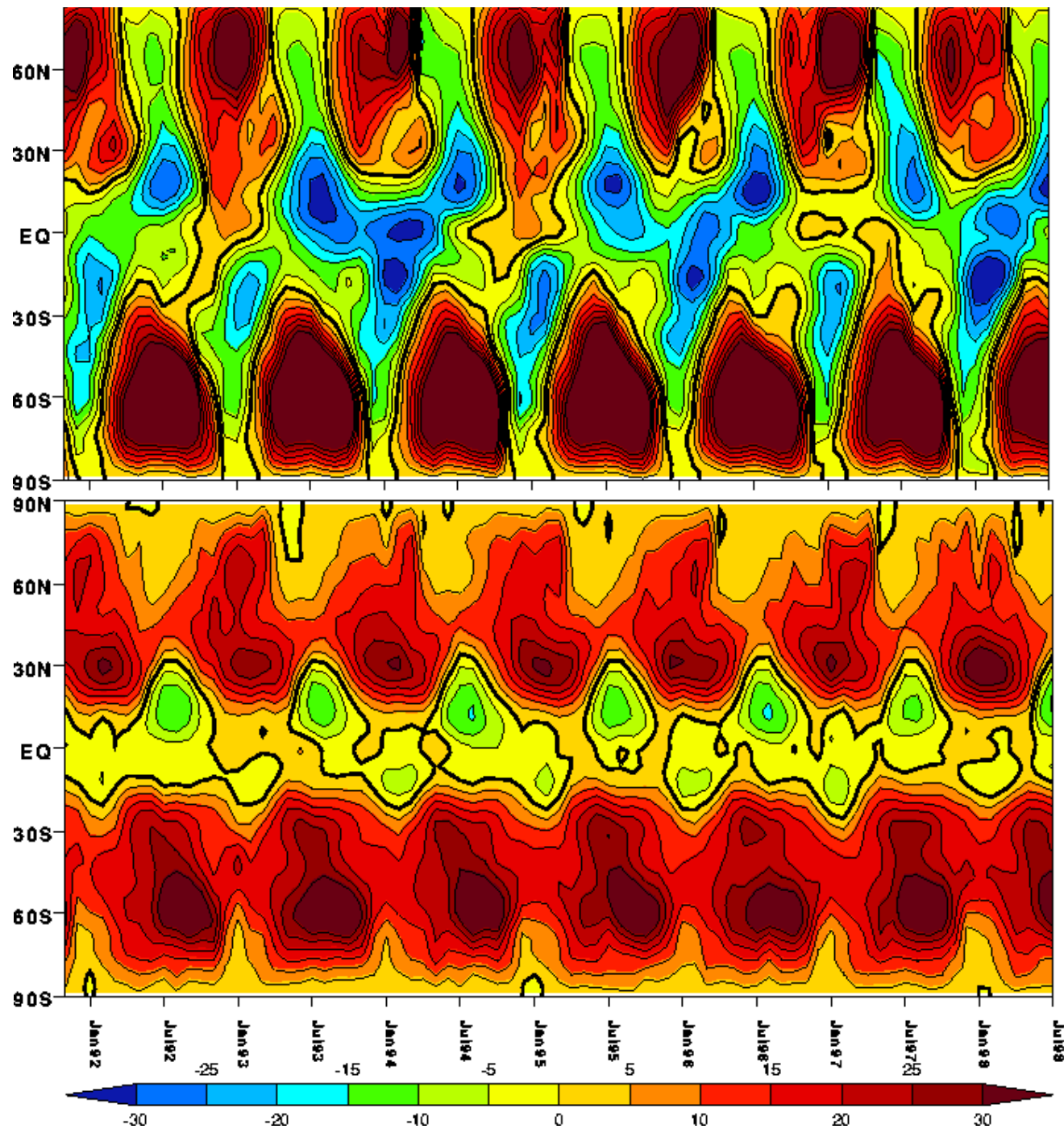
QBO's characteristics



6. Westerlies propagate downward faster than easterlies⁷
7. QBO amplitude peaks around 20hPa, at 40-50 m/s, and generally decreases at lower altitudes
8. Transition from westerlies to easterlies slows down between 30-50hPa (~20-23km)
9. Significant fluctuations in QBO periods and amplitudes.

QBO is strong
near the equatorial
region, at 10hPa

QBO is weaker at
100 hPa



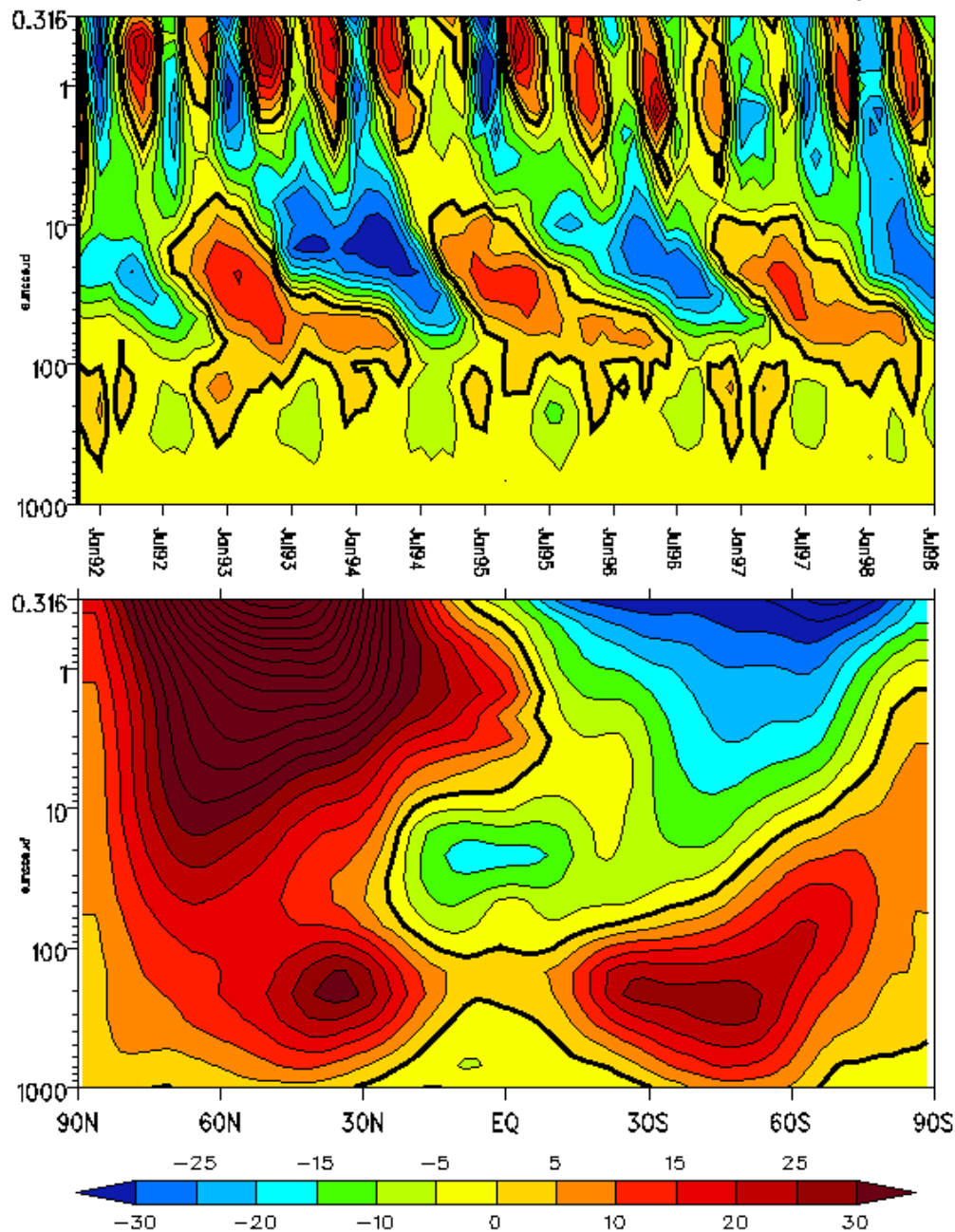
<http://ugamp.nerc.ac.uk/hot/ajh/qbo.htm>

Nov 1991

UKMO Assimilated data

Zonal mean u

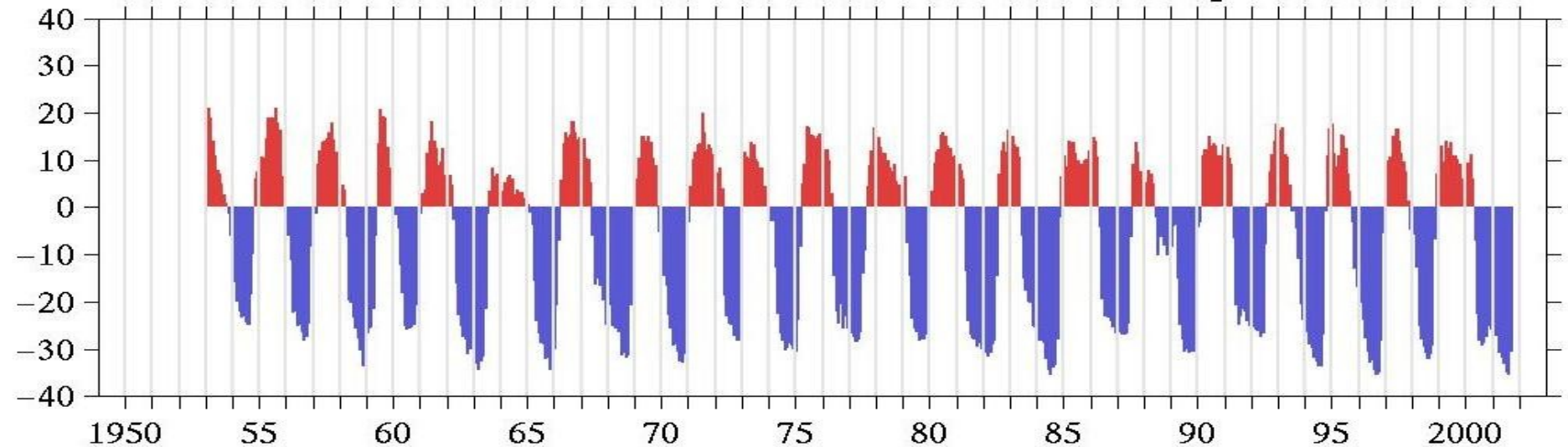
Latitude 1.25 degrees



QBO index

- QBO index at a given altitude: the zonal wind value at that altitude (usually taken at 30hPa, 50hPa, 70hPa)

QBO 30hPa zonal wind (m/s), 1953 – Sep 2001

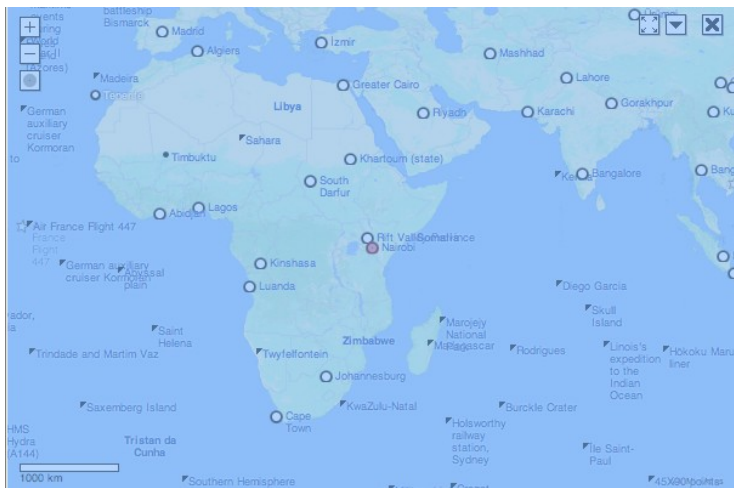


Canton Island (3S,172W) Jan 1953 – Aug 1967
Gan/Maledives (1S, 73E) Sep 1967 – Dec 1975
Singapore (1N,104E) Jan 1976 – Sep 2001

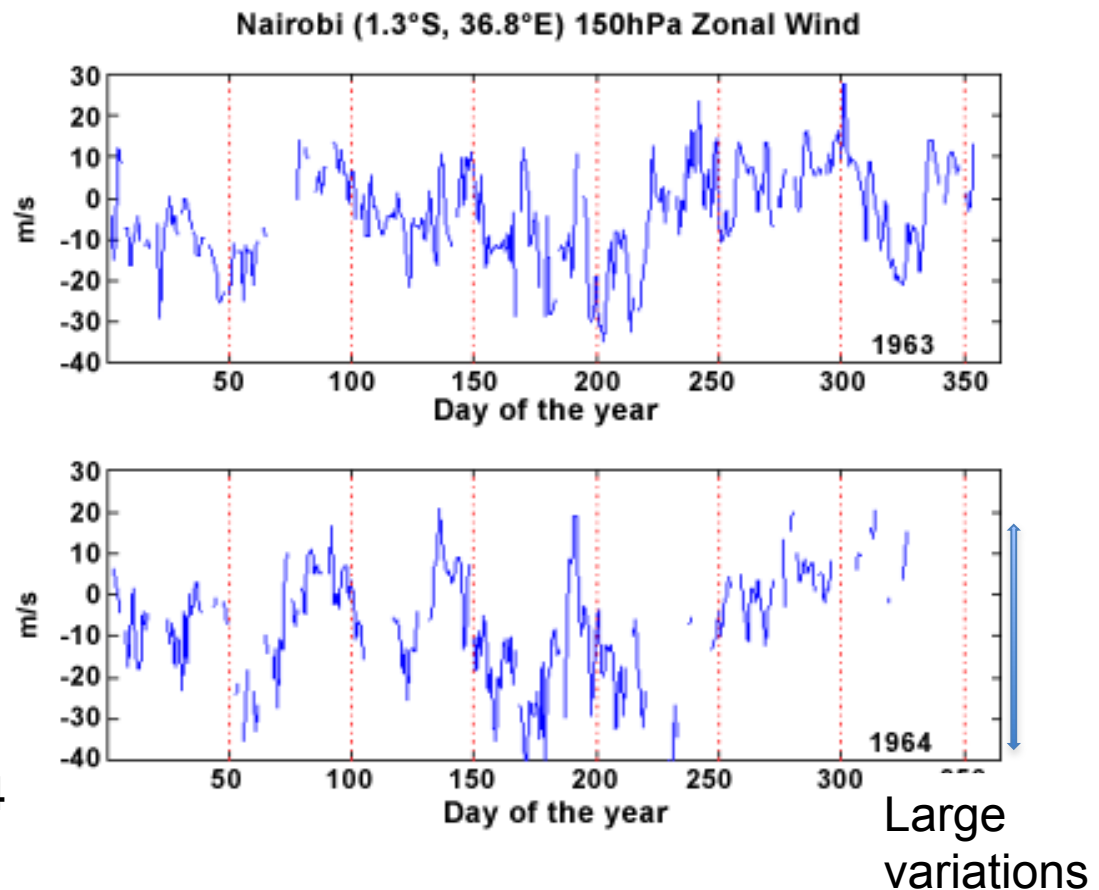
Freie Universitaet Berlin

III. MJO

1971: Madden & Julian found the **40-50 day oscillation** when analyzing zonal wind anomaly in the tropics of the Pacific

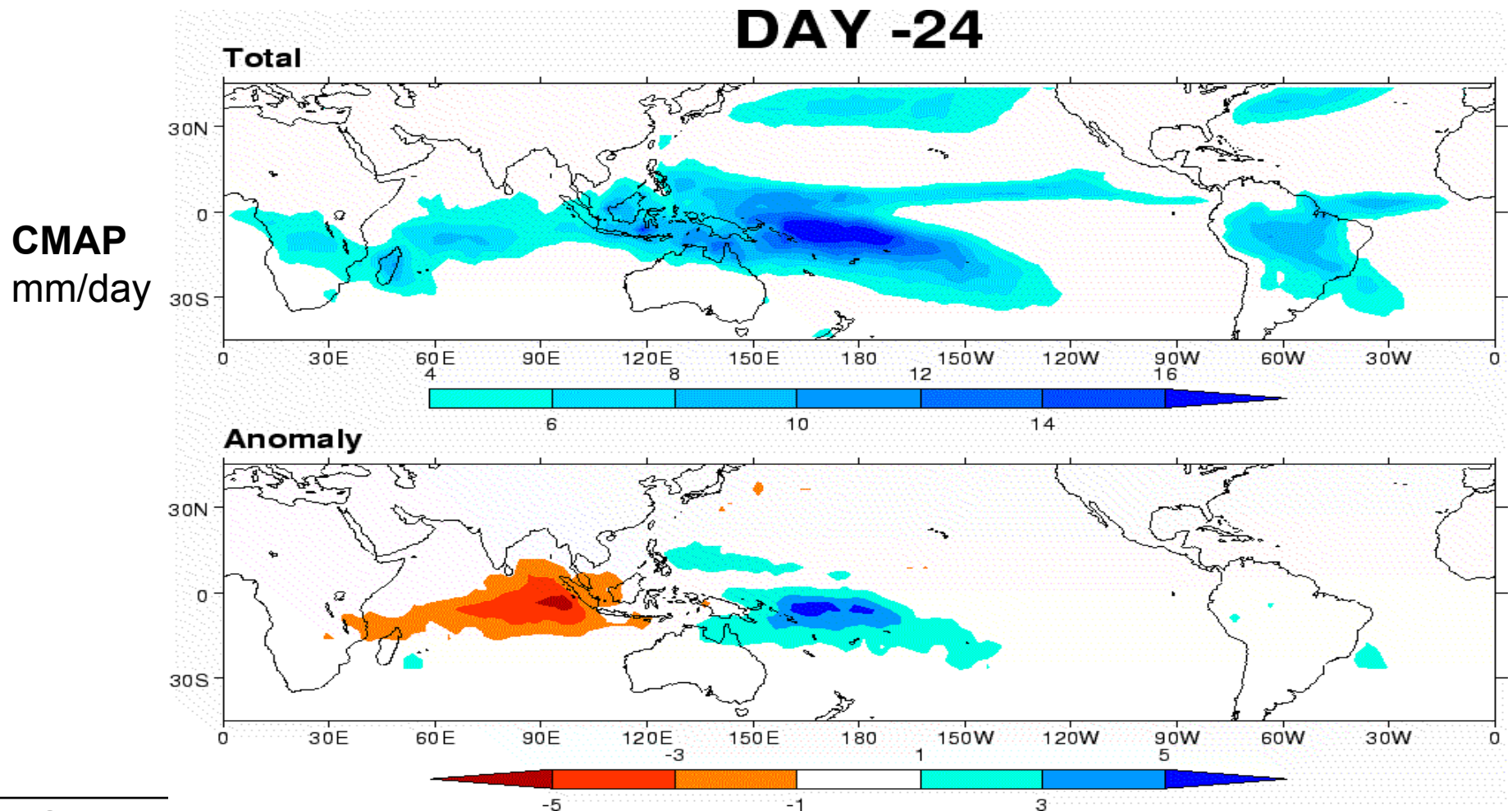


Zonal wind at 150hPa, Nairobi,
after removing seasonal cycles; ,
balloon observation in 1963 & 1964



MJO's characteristics

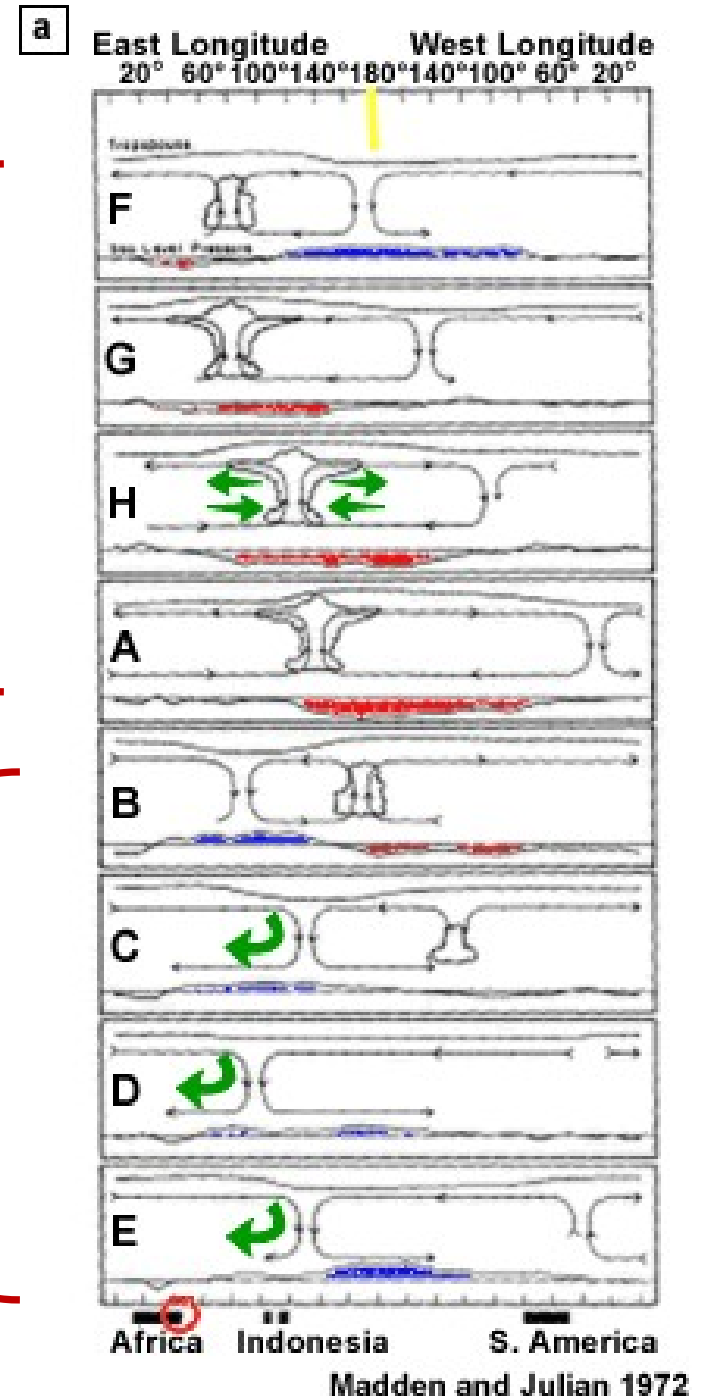
- A large eastward-moving zone of clouds, rain, & related factors (wind, OLR, pressure) along the equator every 30-60 days.
- Most visible in Northern Hemisphere late autumn and winter



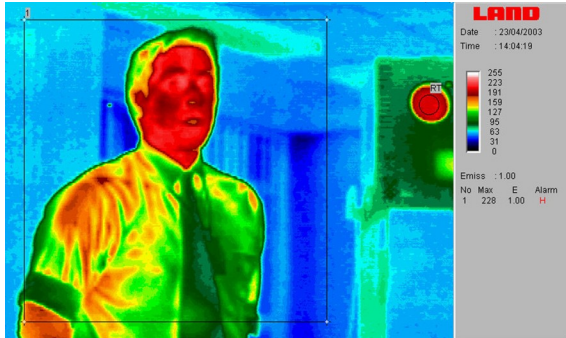
- Time interval for the figures: 5 days.
- negative pressure anomalies (in red) → low pressure → enhancing convection
- Q: Propagation speed?

Wet phase
(active phase)

Dry phase



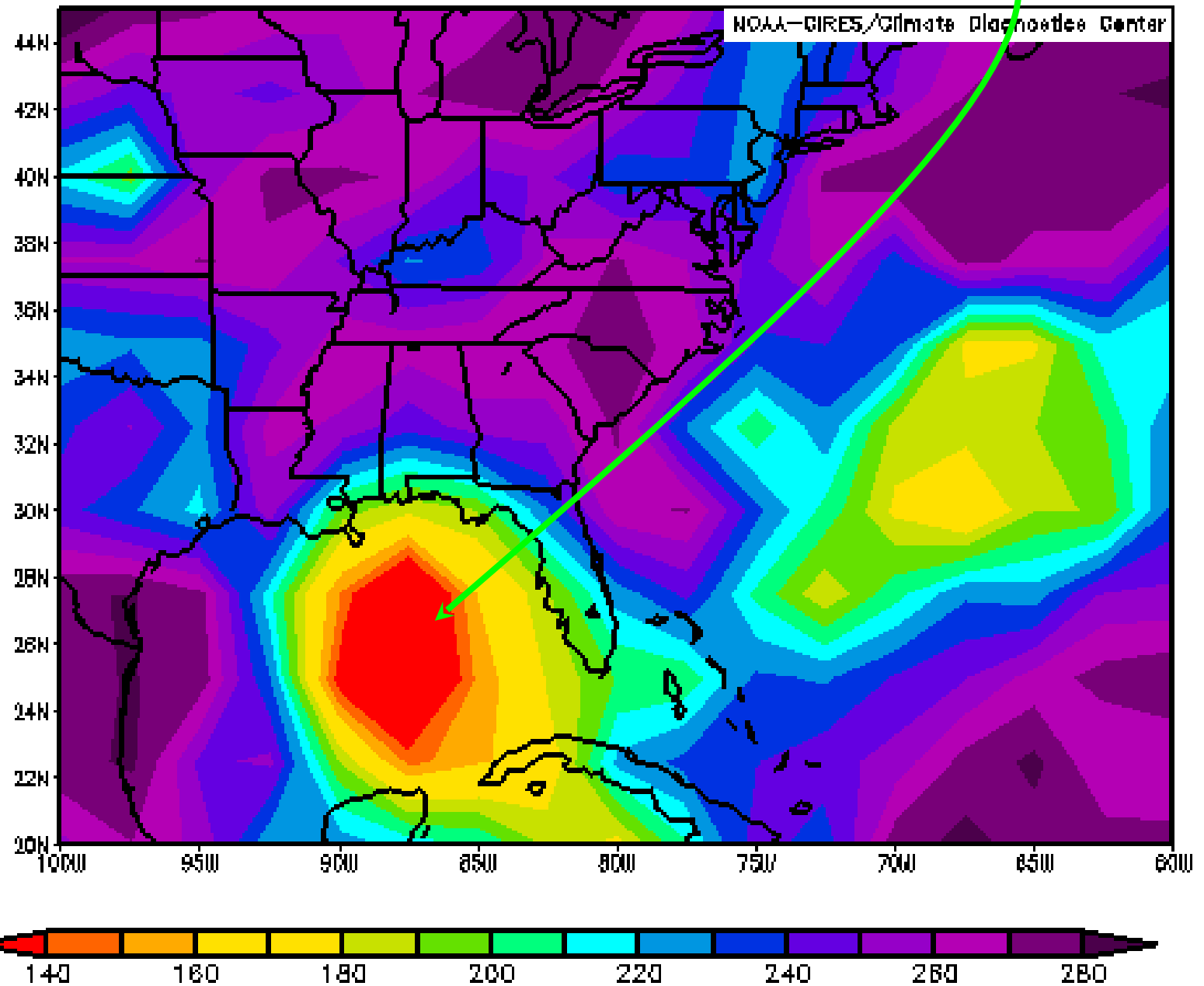
MJO via OLR (Outgoing longwave radiation)



From satellite

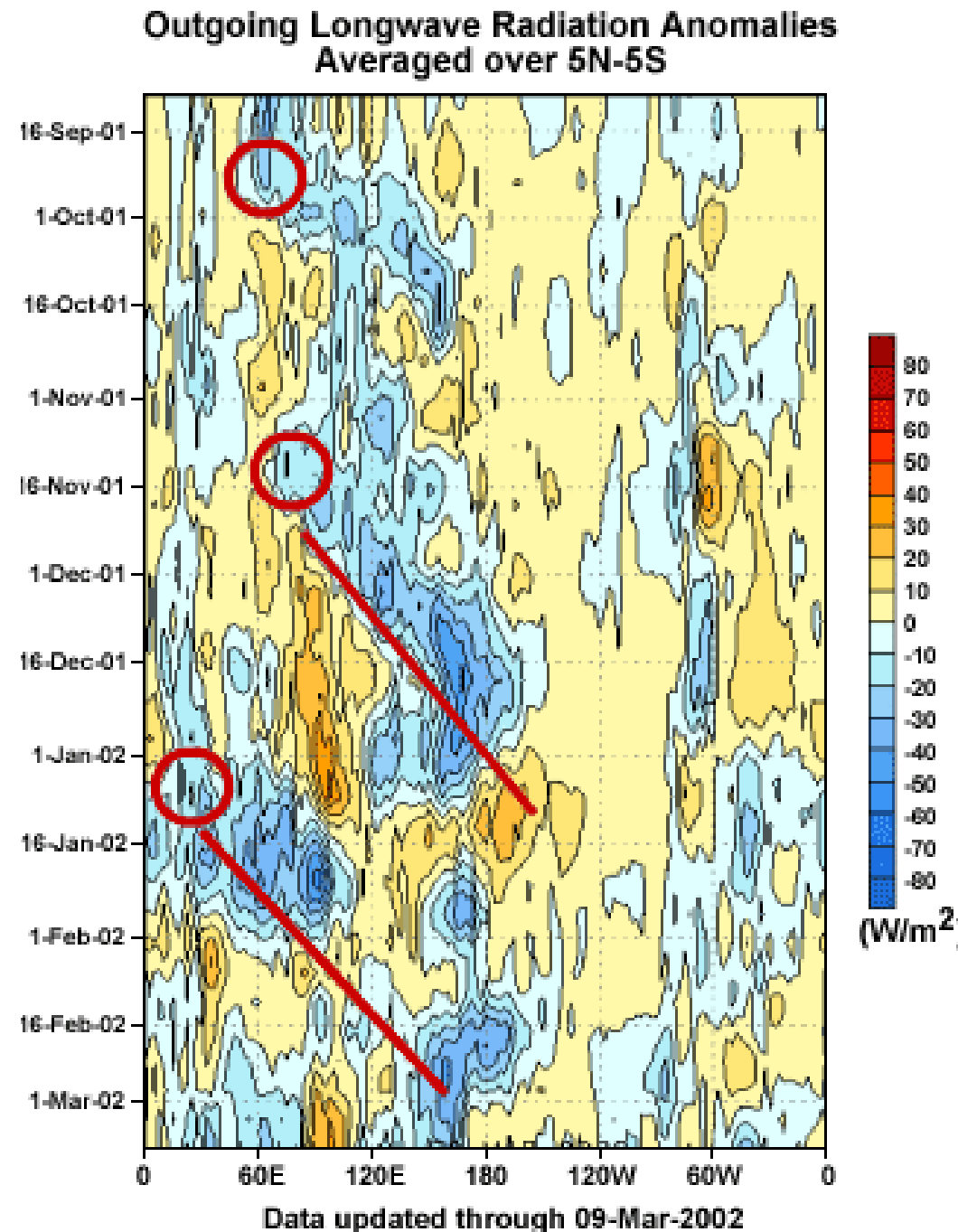


Katrina



Hovmöller diagram

Ernest Aabo Hovmöller (1912-2008)



OLR Anomalies →
movement of convective
regions from West to East

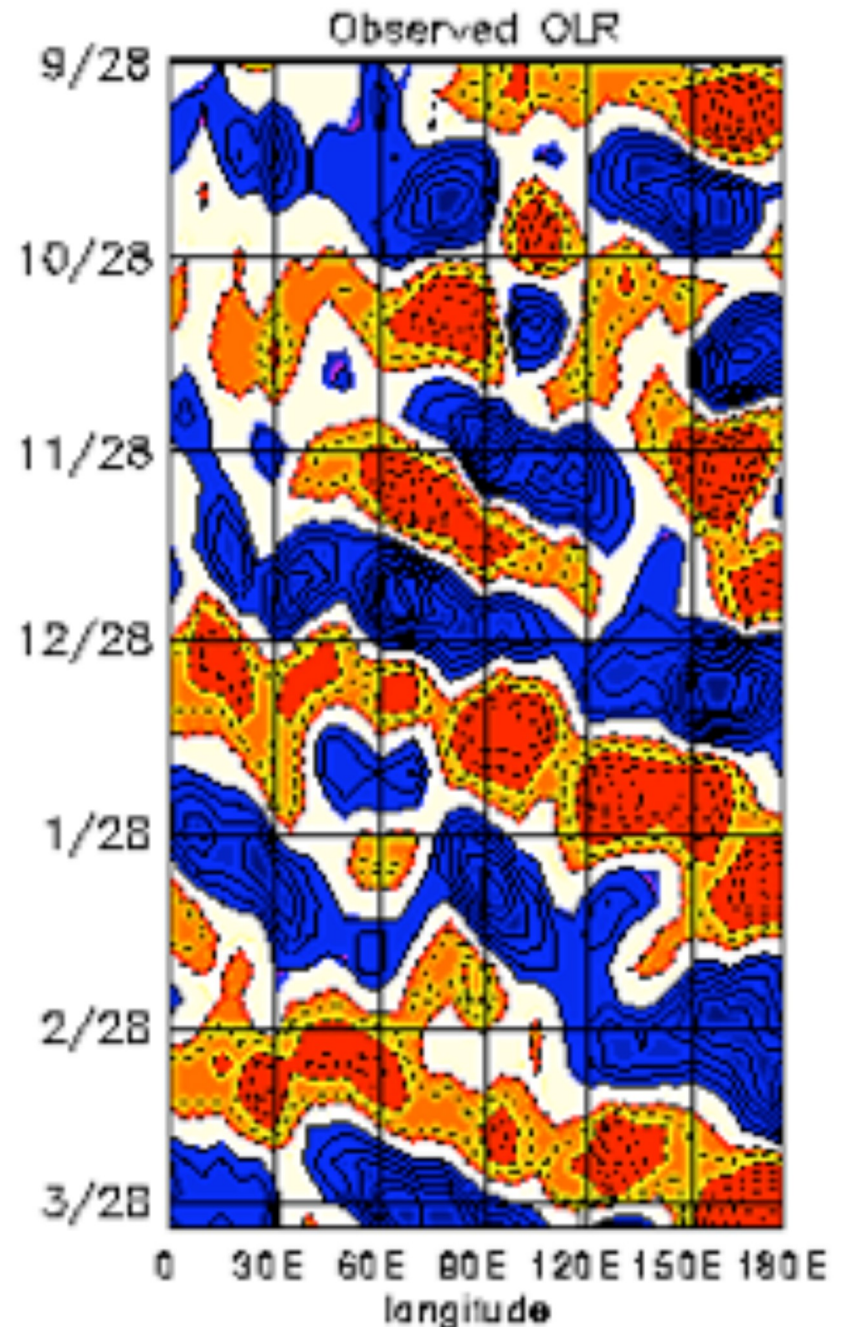
Warm color: positive

Blue: negative

(Geerts và Wheeler (1985))

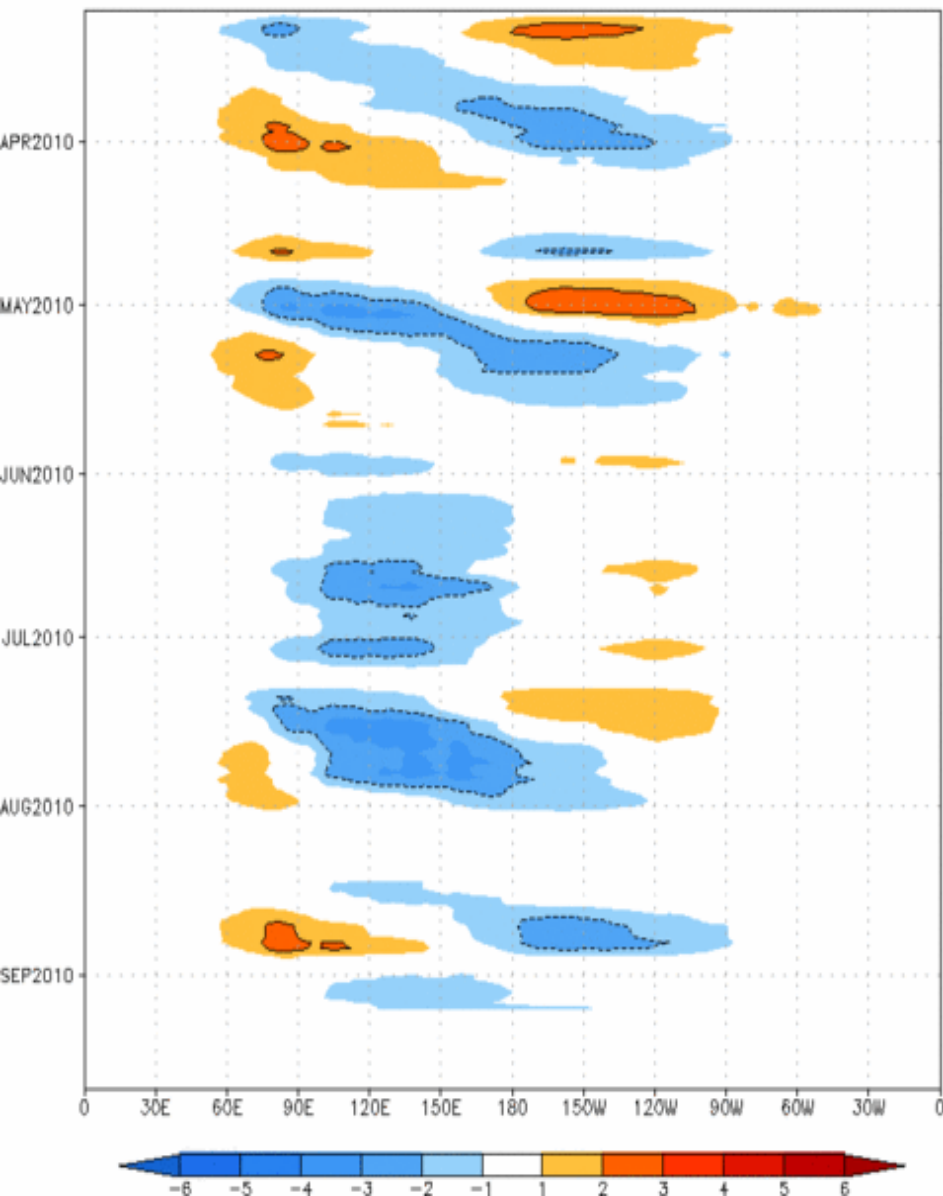
Period?

Movement speed?



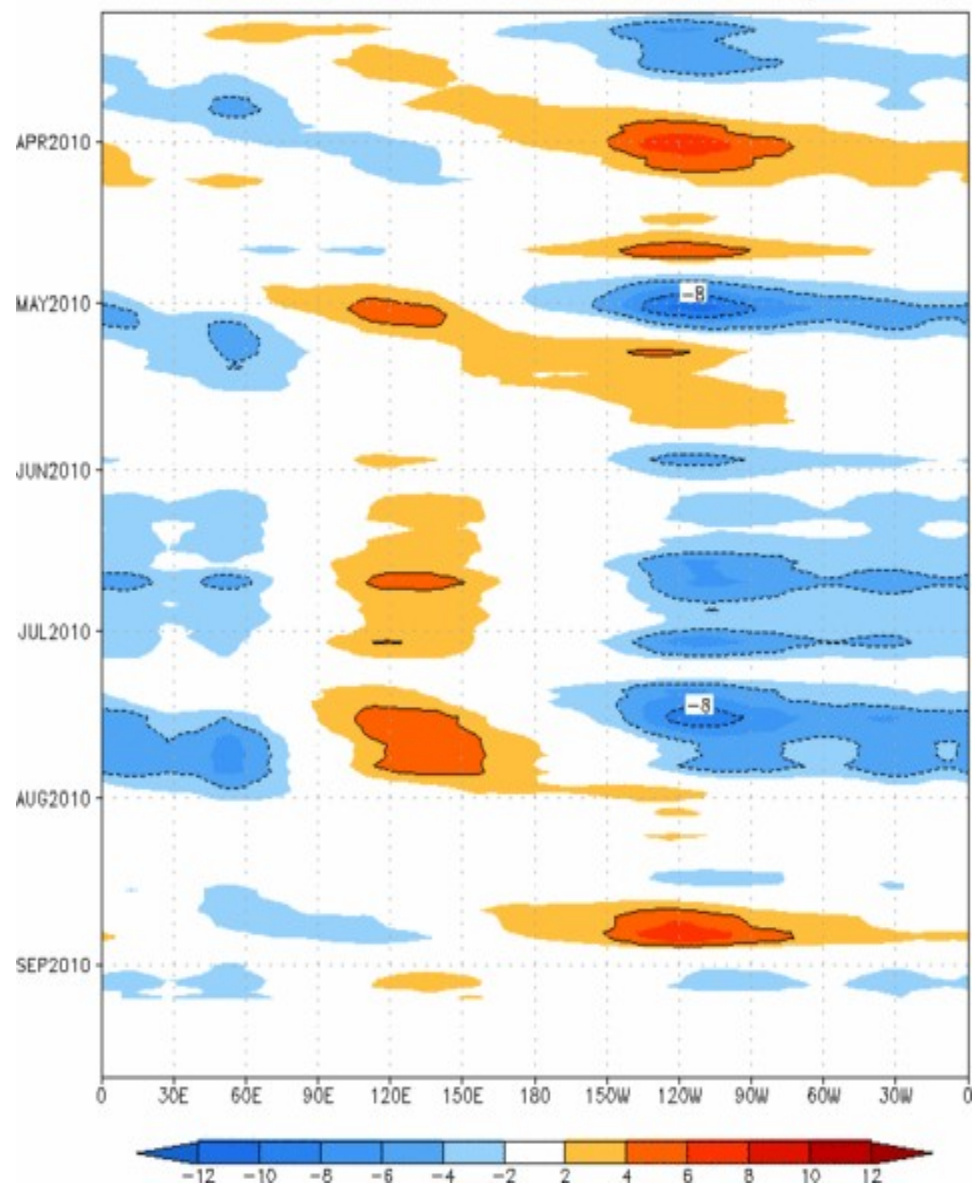
850 hPa Zonal Wind

Reconstructed anomaly field associated with the MJO using RMM1 & RMM2
u850 [7.5°S,7.5°N] (cint:1ms⁻¹) Period:08-Mar-2010 to 07-Sep-2010
The unfilled contours are GEFS forecast reconstructed anomaly for 15 days

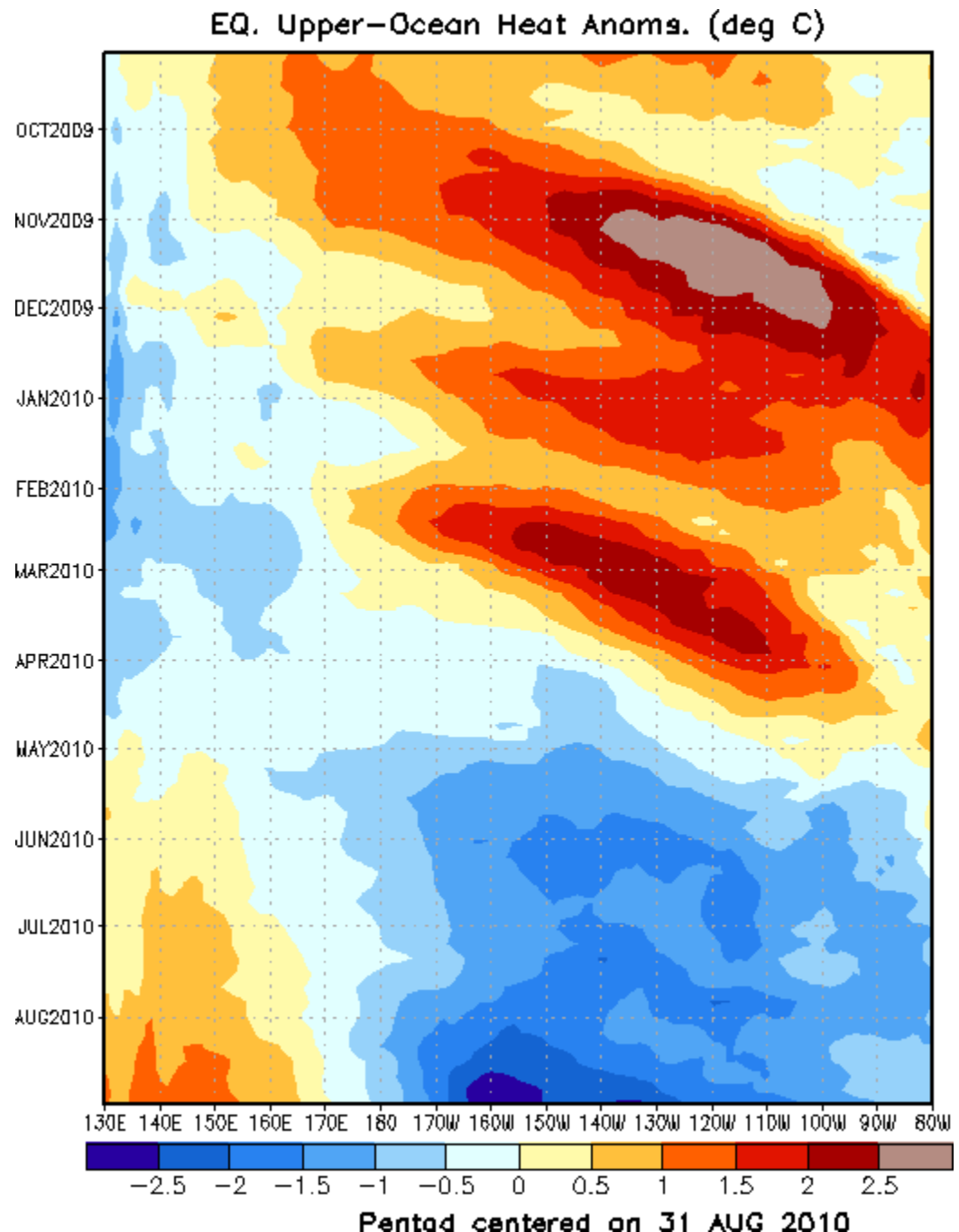


200 hPa Zonal Wind

Reconstructed anomaly field associated with the MJO using RMM1 & RMM2
u200 [7.5°S,7.5°N] (cint:2ms⁻¹) Period:08-Mar-2010 to 07-Sep-2010
The unfilled contours are GEFS forecast reconstructed anomaly for 15 days

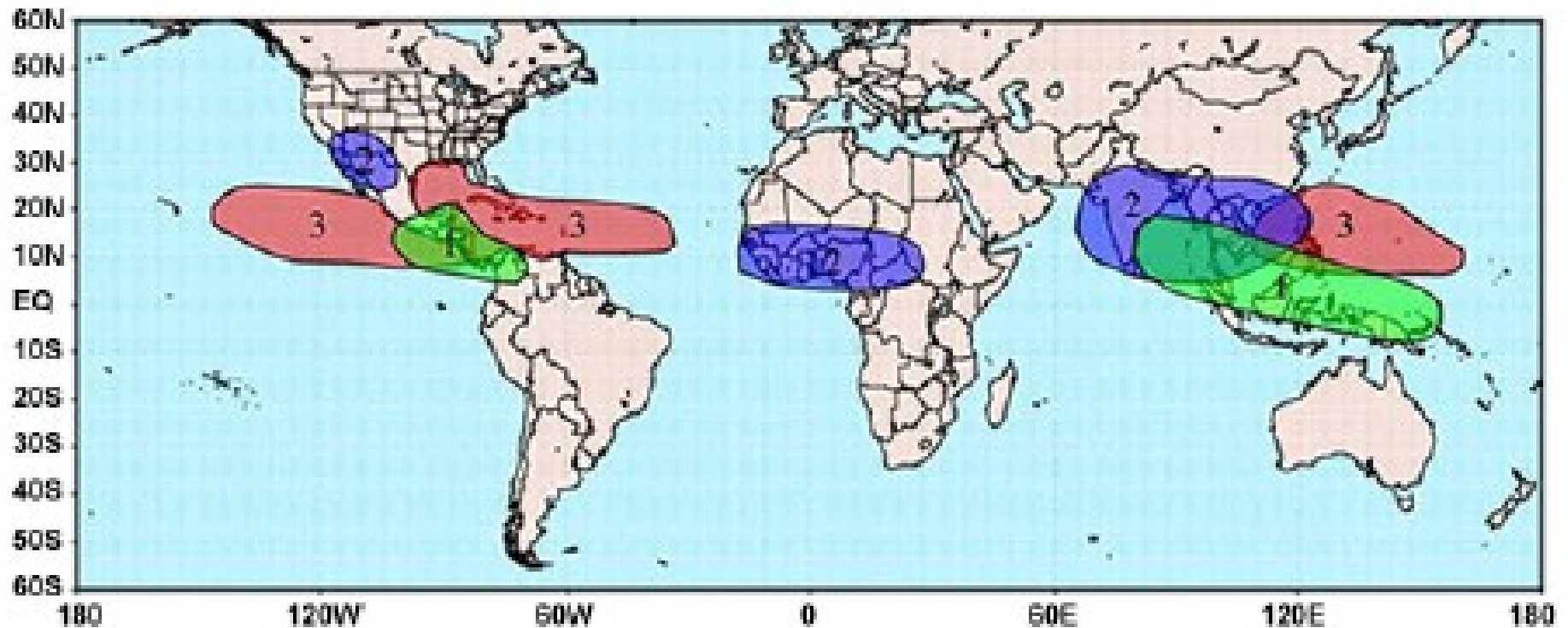


0-300m ocean temperature anomaly



Impact of MJO

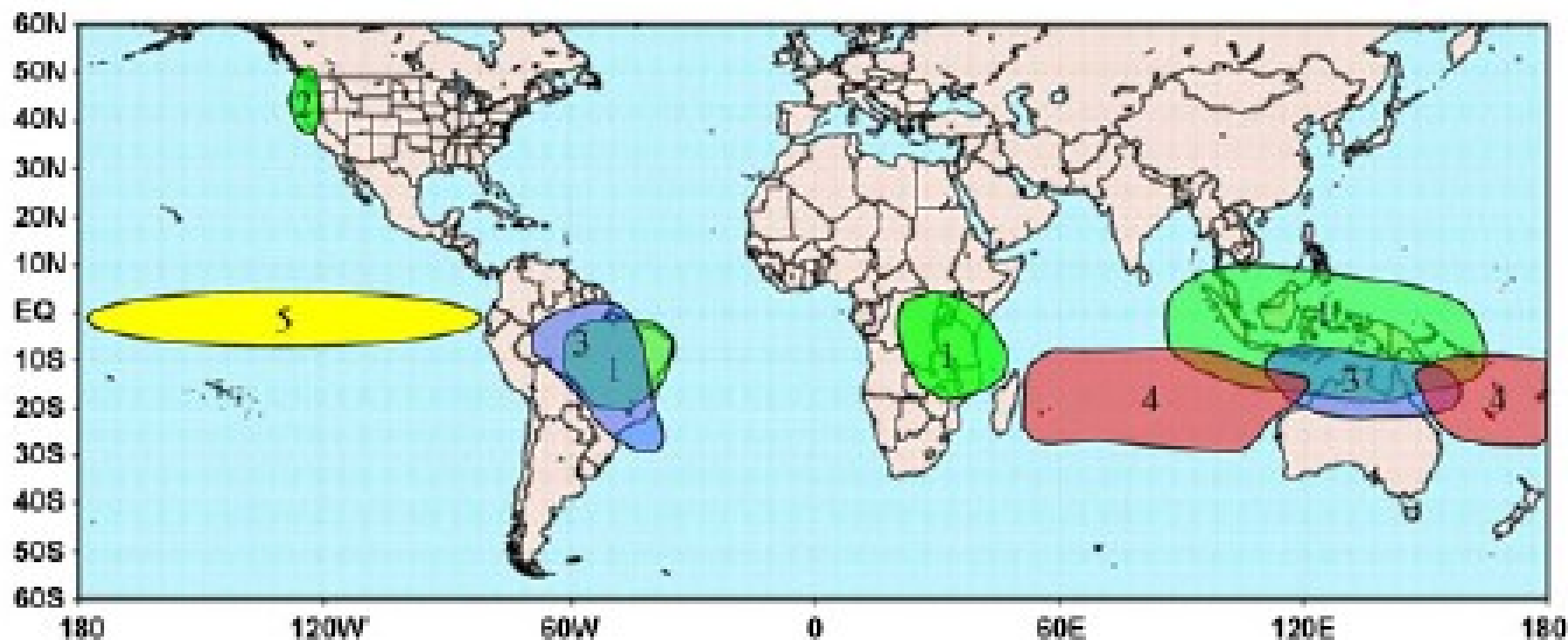
MJO Impacts during Boreal Summer



- 1 - Alternating wet/dry conditions**
- 2 - Modulation of the monsoon**
- 3 - Modulation of tropical cyclone activity**

NOAA / CPC / NCEP / NWS

MJO Impacts during Boreal Winter



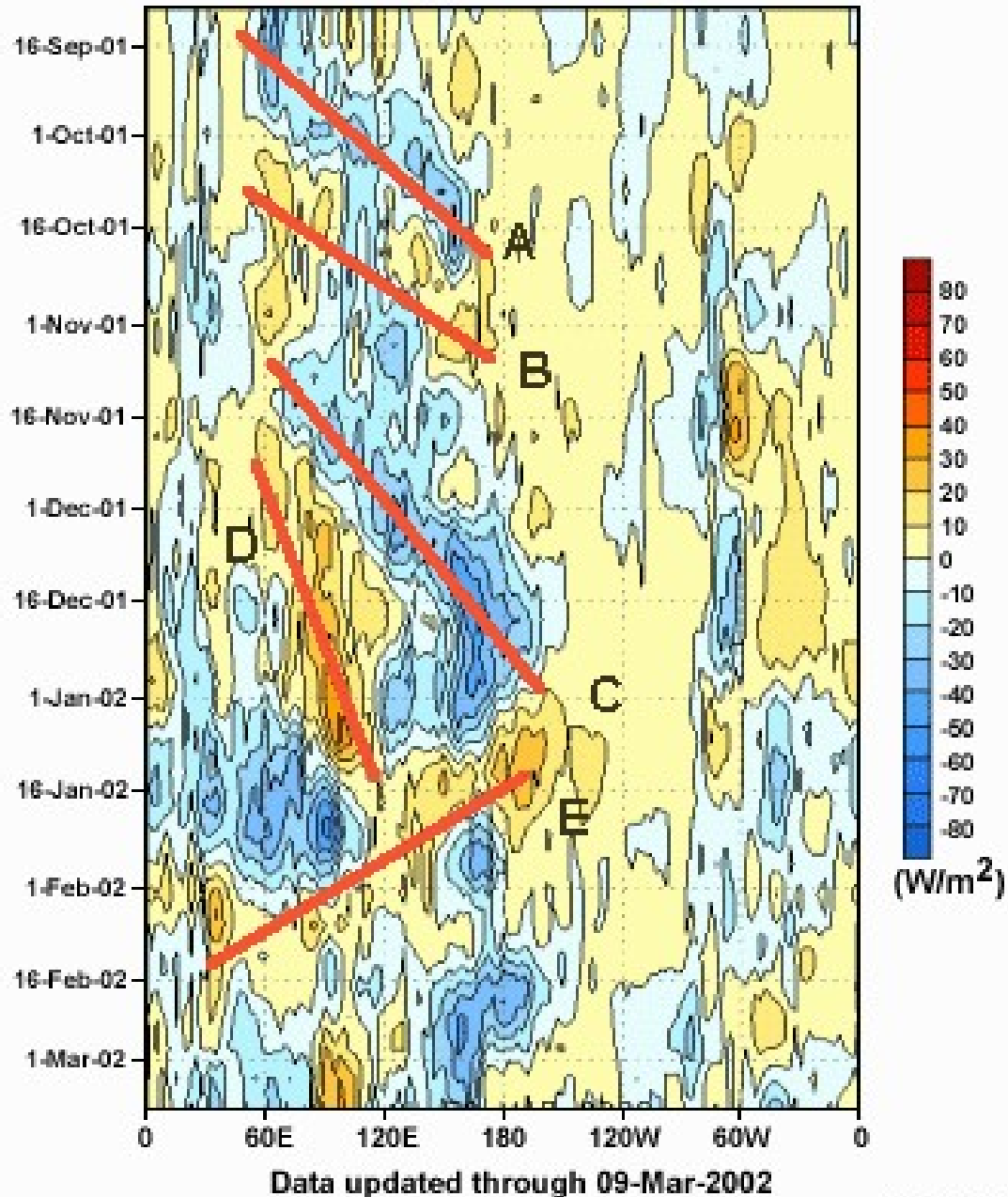
- 1 - Alternating wet/dry conditions
- 2 - Tropical moisture plume to higher latitudes
- 3 - Modulation of monsoon systems
- 4 - Modulation of tropical cyclone activity
- 5 - Modulation of ENSO through oceanic Kelvin waves

NOAA / CPC / NCEP / NWS

MJO Monitoring and Prediction

NOAA National Weather Service (NWS) Climate Prediction Center (CPC) Daily MJO Indices	http://www.cpc.noaa.gov/products/precip/CWlink/MJO/mjo.shtml http://www.cpc.ncep.noaa.gov/products/precip/CWlink/MJO/index.primjo.html
NOAA Earth System Research Laboratory Experimental MJO Predictions	http://www.cdc.noaa.gov/mjo/
Australian Bureau of Meteorology tropical variability (MJO and equatorial waves) forecasts	http://www.bom.gov.au/bmrc/clfor/cfstaff/matw/maproom/RMM/ http://www.bom.gov.au/bmrc/clfor/cfstaff/matw/maproom/OLR_modes/index.htm
MJO forecasts by Paul Roundy, State University of New York at Albany (SUNY, Albany)	http://www.atmos.albany.edu/facstaff/roundy/waves/
MJO forecasts by Barney Love and Adrian Matthews University of East Anglia	http://envaml.env.uea.ac.uk/mjo_forecast.html

Outgoing Longwave Radiation Anomalies Averaged over 5N-5S



Quizz

- ORL anomalies
- Which lines correspond to an active MJO phase?

Practice #11.1: ENSO

- file sst.mnmean.nc
- <https://www.esrl.noaa.gov/psd/data/gridded/data.noaa.oisst.v2.html>
- Calculate the Nino3 index and identify the El Niño and La Niña years for the period 1981-present.
 - ✓ *Hint #1: Niño3 area: 5S-5N and 150W-90W.*
 - ✓ *Hint #2: criteria: ENSO occurs when the 3-month running average of SST anomalies exceeds 0.5°C for five consecutive 'months'*

Practice #11.2: QBO

Download QBO data from <http://www.geo.fu-berlin.de/en/met/ag/strat/produkte/qbo/>

Qbo.dat

Singapore.dat

**QBO Monthly mean zonal wind components (0.1 m/s)
at Singapore (48698), 1N/104E**

Questions:

- 1) From the data downloaded, prove that QBO exists
- 2) Identify QBO's characteristics from the data

	2010											
hPa	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
10	-171	-282	-259	-230	52	128	140	150	149	163	144	59
12	-280	-392	-370	-348	-31	123	136	157	148	155	161	8
15	-362	-347	-332	-342	-236	59	140	150	153	140	147	39
20	-330	-309	-317	-326	-347	-104	92	111	134	111	116	30
25	-290	-300	-301	-297	-295	-209	56	112	121	112	119	83
30	-265	-272	-269	-272	-263	-285	-59	117	134	116	118	83
35	-212	-189	-237	-254	-252	-289	-159	78	133	113	111	87
40	-98	-103	-185	-251	-253	-278	-276	0	120	102	107	89
45	-21	-24	-90	-236	-252	-270	-289	-72	102	90	94	86
50	58	51	-8	-159	-234	-235	-278	-178	38	77	72	76
60	101	96	64	0	-158	-180	-209	-259	-164	16	57	66
70	82	61	49	56	-56	-137	-163	-163	-185	-37	41	44
80	60	-9	19	45	7	-134	-143	-98	-141	-78	-16	23
90	48	-55	-23	41	-9	-136	-160	-100	-150	-119	-84	2
100	3	-53	-27	24	-49	-136	-159	-144	-169	-125	-125	-87

Practice #11.3*(bonus): MJO

- Using GPCP global rainfall data to detect MJO for a period of any 2 years of your choice; and to show MJO's characteristics.

GPCP: <https://climatedataguide.ucar.edu/climate-data/gpcp-daily-global-precipitation-climatology-project>

- From the SST data, calculate the Nino3 index and identify the El Niño and La Niña years for the period 1981-present.
- Niño3 area: 5S-5N and 150W-90W.
- ENSO criteria: five consecutive 3-month running mean SST anomalies exceed the threshold (0.5°C)

```

1  # need to install nc-time-axis to handle the time axis
2  !pip install nc-time-axis

1  import netCDF4 as nc
2  import numpy as np
3  import matplotlib.pyplot as plt
4  import cftime
5  import nc_time_axis
6  # ===== #
7  # Define function to estimate the moving average of a data series
8  def calc_mov_avg (dat_in, mov_steps):
9      steps = np.ones(int(mov_steps))/float(mov_steps)
10     return np.convolve(dat_in, steps, 'same')
11 # ===== #
12 # Define function to estimate the climatology of a monthly data series
13 def calc_climato (dat_in):
14     n = len(dat_in)
15     nyr = int(np.ceil(n/12.0)) # number of years in the data series
16     dat_expanded = np.r_[dat_in, np.zeros(12*nyr-n)]
17                     # full nyr*12 elements
18     dat_2_divide = dat_expanded.reshape(-1,12).sum(0)
19                     # create matrix of 12 columns, nyr lines,
20                     # then sum up for eachh columnn
21     idx = np.r_[np.ones(n), np.zeros(12*nyr-n)]
22     idx_divisor = idx.reshape(-1,12).sum(0)
23     climato_12 = np.divide(dat_2_divide, idx_divisor)
24                     # 12 climatological values for each month
25     climato_fnl = np.tile(climato_12,nyr).flatten()[:n]
26                     # duplicating the climatological values
27     return climato_fnl
28 # ===== #
29 # Read input file from the NOAA SST monthly dataset
30 infile="sst.mnmean.nc"

```

```

30 infile="sst.mnmean.nc"
31 f = nc.Dataset(infile, 'r')
32 # Get variables
33 lons=f.variables['lon'][:]
34 lats=f.variables['lat'][:]
35 times=f.variables['time'][:]
36 xtimes=nc.num2date(f.variables['time'][:],units=f.variables['time'].units)
37 sst3d=f.variables['sst'][:]
38 f.close()
39 # ===== #
40 #select NINO regions
41 latidx_nino3 = (lats >=-5. ) & (lats <=5. )
42 lonidx_nino3 = (lons >=210.) & (lons <=270.)
43 # Time series of average SST over the NINO regions
44 sst3 = np.mean(sst3d[:, latidx_nino3] [..., lonidx_nino3] ,axis=(1,2))
45 # Estimate anomaly for each NINO region
46 nino3 = sst3 - calc_climato(sst3)
47 # Estimate the 3-month moving average for NINO3 SST anomaly
48 nino3_mov = calc_mov_avg(nino3,3)
49 # ===== #
50 # Plot Figure
51 fig,(ax1,ax2)=plt.subplots(2,1,sharex=True,figsize=(7,6))
52 ax1.plot(xtimes,nino3 , color = "green",lw = 1.5, label = "Niño 3" )
53 ax1.set_ylabel('(°C)',fontsize=14)
54 ax1.set_title('SST anomalies for the Niño3 region (°C)')
55 ax1.legend(loc='upper right')
56 ax1.grid(True,dashes=(1,2,1,2))
57
58 ax2.plot(xtimes,nino3_mov, color = "black", lw = 2)
59 ax2.fill_between(xtimes,nino3_mov,0, where= nino3>0, color='red')
60 ax2.fill_between(xtimes,nino3_mov,0, where= nino3<0, color='blue')
61 ax2.axhline(y=0.5,color="red",linestyle='--')
62 ax2.axhline(y=-0.5,color="blue",linestyle='--')
63 ax2.set_ylabel('(°C)',fontsize=14)
64 ax2.set_xlabel('Year',fontsize=14)
65 ax2.set_title('3-month moving average of SST anomaly of the region Niño3 (°C)')
66 plt.grid(True,dashes=(1,2,1,2))
67 # ===== #
68 # END

```